PVMouse 3.0: Design and Development Path for a Biological World Model

Multiscale Modeling and Intervention Prediction, Starting with a Virtual Mouse

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Readers can select chapters according to their interests. Chapters 4, 5 and 8 address complementary questions: what biological information the model represents, how that information is used in computation, and how the system will be developed in stages.

Chapter 1 — Motivation and Vision
Why move from studying individual biological processes toward predicting responses across the whole organism?

Chapter 2 — Biological Foundations
Which biological observations support the four working hypotheses for modeling living systems?

Chapter 3 — Biological State Space
How can a biological state space organize observations from different studies and connect state estimation with response prediction?

Chapter 4 — Biological Architecture
What biological information should be represented at the molecular, cellular, spatial organ and multi-organ levels?

Chapter 5 — Computational Framework
How are these levels connected, trained and used for prediction, and how will their predictive value be tested?

Chapter 6 — Continual Learning and Experimental Feedback
How can new data and experimental results improve the model and inform the selection of subsequent experiments?

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How can a shared model support five types of research on disease and intervention?

Chapter 8 — Development Path
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Chapter 9 — Outlook and Conclusions
How can this research approach extend to broader biological contexts and longer-term goals?

Abstract

Predicting multi-organ responses and organism-level functional outcomes from local molecular changes requires a framework that connects dispersed biological observations to testable system-level predictions. This white paper proposes PVMouse 3.0 (Panoramic Virtual Mouse V3.0), a minimum viable product (MVP) for a biological world model, starting with C57BL/6J mice. It uses a biological state space to organize multiscale observations and link state estimation, intervention responses and experimental feedback.

The proposed architecture integrates molecular priors, cell state representations, spatial organ models and a dynamic multi-organ graph model. Molecular knowledge guides cell state learning, while single-cell representations, cell composition and spatial organization together define regional and organ-level states. Graph message passing and neural dynamics modeling aim to capture inter-organ coupling and state changes over time and in response to interventions. Joint training on multi-tissue transcriptomic data and physiological, biochemical and pathological readouts from the same animal aims to constrain these relationships and link molecular changes to functional outcomes.

PVMouse 3.0 aims to jointly predict transcriptomic changes and associated phenotypes in the brain, heart, liver, skeletal muscle and spleen under specified conditions of genetic background, sex, age and gene editing. Development will proceed from reconstructing normal states to modeling genetic perturbations and temporal dynamics. Cross-sectional observations across ages will provide a population-level physiological reference, while perturbation data will inform candidate trajectories under defined initial conditions and explicit dynamical assumptions. These trajectories will be calibrated using repeated longitudinal functional measurements and results from independent experiments.

The central test is whether cellular, spatial and multi-organ information improves predictions of molecular responses and functional changes under different perturbations. Initial development and testing will use single-gene knockouts and matched controls to build and test the model, evaluating missing observation estimation and intervention consequence prediction respectively. Independent data, simple baselines and follow-up experiments will assess the contributions of information sources and model components. Prediction errors will guide additional observations and model refinement.

Initial public datasets have been collected, some algorithmic components have been built and data cleaning is underway; the full system remains to be completed and validated. This white paper invites methodological discussion, data and model collaboration, and joint experiments.

Chapter 1 | Biology is not dying. Our way of doing biology is.

It is the best of times, and perhaps the most challenging of times, for biology.

Over the past two decades, our ability to observe, perturb, and computationally study living systems has grown at an unprecedented pace. From microarrays to bulk RNA-seq, from single-cell transcriptomics to spatial omics, and from CRISPR gene editing to AI-driven structure prediction, we can now observe biological activity in greater depth than ever before. Yet the growth in our ability to observe has not been matched by growth in our ability to understand. This gap is becoming increasingly apparent in our ability to interpret data, integrate knowledge and develop theories:

- We collect more and more data, yet still struggle to turn them into a systems-level explanation of complex biological processes.
- We publish more and more papers, yet our understanding of how biological systems work as a whole advances much more slowly.
- We discover more and more correlations, yet we are still far from turning them into theories that reliably predict how biological systems respond.

Therefore, what truly faces a crisis is not biology itself, but the research paradigm that has driven biology over recent decades. If we continue to use traditional approaches to address future questions in the life sciences, we may have no shortage of data, papers or experimental results, yet find it increasingly difficult to truly understand living processes. In other words:

Biology is not dying.

Our way of doing biology is.

1.1 Why classical biology has been so successful

Modern biology has a remarkable record of discovery and innovation, spanning the DNA double helix, the central dogma, classical genetics, recombinant protein therapies, genetic engineering, molecular tumor classification and precision medicine. One representative category of success has come from identifying key drivers of disease and using experiments to establish how they shape biological traits. Research on thalassemia (*Orkin et al., 1982*), cystic fibrosis (*Riordan et al., 1989*), familial hypercholesterolemia (*Brown & Goldstein, 1986*) and chronic myeloid leukemia (*Druker et al., 2001*) illustrates the power of this approach. In each case, researchers can gradually build causal explanations linking phenotypes around genes, protein functions, and related biological processes. Such systems, in which key factors can be identified and experimentally manipulated, have allowed the controlled variable method to play a central role.

Modern experimental biology is built on an extremely successful methodological foundation: Observation → Formulate Hypothesis → Control Variables → Experimental Validation. For decades, this approach has driven major advances in biology, establishing itself as the gold standard for research in the life sciences.

However, the very problems biology seeks to understand are changing.

1.2 The remaining problems are no longer of the same kind

Today, major challenges in the life sciences include aging, complex diseases such as obesity, type 2 diabetes, metabolic dysfunction-associated steatotic liver disease, cardiovascular disease, neurodegenerative disease and autoimmune disease, and clinical problems such as cancer drug resistance and relapse. These challenges involve multiple genetic and environmental factors whose effects are shaped by intercellular communications, multi-organ cross-talk, and temporal dynamics (*Hasin et al., 2017*). In these contexts, living systems are better understood as high-dimensional dynamic networks than as simple chains of cause and effect.

This shifts the focus of research. Identifying key factors that cause disease remains essential, but we also need to understand how multiple factors work together to push a biological system into a disease state and sustain that state.

1.3 Experimental complexity is outgrowing experimental capacity

Biology has always been an experimental science. One of its central tasks is to establish causal relationships between interventions and biological responses. Manipulating a factor of interest while holding other variables constant and using appropriate controls is one of the most established and direct ways to do this.

In complex disease research, however, interactions among many factors make it difficult to test every possible combination or extrapolate results from one set of conditions to another. This does not diminish the ability of controlled experiments to establish causal effects. Even with just 20 genes, each with two possible states, testing every combination would require more than one million experimental conditions. Gene expression, protein activity and metabolic activity also vary continuously in living systems, making the full range of possible responses even harder to explore.

Experiments can still provide reliable causal knowledge. However, testing one condition at a time is becoming an increasingly difficult way to systematically characterize the effects of multiple factors in complex diseases.

1.4 After the omics revolution, interpretation remains a bottleneck

Over the past few decades, the life sciences have undergone an unprecedented revolution in our ability to observe biological systems. Advances span sequencing-based genomics, transcriptomics and microbiome profiling; mass spectrometry-based proteomics, metabolomics and lipidomics; and histology-integrated spatial omics. Today, a single omics experiment can generate more data than an entire laboratory could produce in a year two decades ago. However, the accumulation of vast amounts of observational data has not automatically led to a more integrated understanding of living systems.

Many studies now follow a familiar workflow: DEG analysis → functional enrichment analysis (including Gene Ontology and pathway analyses) → validation of selected candidate genes → mechanistic studies. There is nothing inherently wrong with this approach, but it is becoming less able to answer questions about complex systems as a whole.

1.5 Hypothesis-driven research is approaching its limits

Hypothesis-driven research has long been a foundation of modern experimental biology. At my PhD defense in 2018, I framed my study around a research hypothesis: first propose a hypothesis, then design experiments to test it. This approach works well when causal relationships are relatively clear. But in complex diseases shaped by many interacting factors, researchers face a more fundamental difficulty before experiments even begin: deciding which of the many possible hypotheses is most worth testing. Even when experiments support a hypothesis about one part of the system, fitting that finding into an explanation of the system as a whole remains difficult.

The abundance of omics data has not automatically solved the problem of choosing which hypotheses to pursue. Researchers must identify mechanisms worth investigating among the many factors that change together. Redundant regulation and adaptive compensation further complicate this task, making it difficult to understand what any one factor does outside their system context (*El-Brolosy et al., 2019*).

To develop more informative hypotheses, researchers may need to frame their questions differently. As Albert Einstein wrote in *The Evolution of Physics* (*Einstein & Infeld, 1938*), "The formulation of a problem is often more essential than its solution, which may be merely a matter of mathematical or experimental skill. To raise new questions, new possibilities, to regard old problems from a new angle, requires creative imagination and marks real advance in science." The breakthrough of general relativity came from rethinking gravity in terms of the properties of space and time themselves. A similar shift may be needed in the study of complex diseases: extending our questions beyond the effects of individual factors to ask how multiple factors together shape a disease state, and which interventions might change it.

1.6 From individual studies to an integrated understanding of the organism

Millions of biomedical papers are published worldwide each year, making this the largest knowledge-producing system in human history—and one of the most fragmented. Laboratories work with different experimental models, collect different samples at different times, and use different experimental conditions and analytical methods. Their studies reveal different aspects of living systems, but differences in sampling design, levels of observation, time frames and experimental conditions often make the results difficult to integrate. As a result, individual findings do not readily add up to a coherent understanding of the system. Many valuable negative experimental results also remain unknown to other researchers in this distributed system, leading to substantial waste of resources.

We produce biological observations far faster than we integrate biological knowledge. Bringing dispersed data and findings together is essential if individual observations are to contribute to an understanding of the whole organism. This calls for a unified, extensible framework that can continue learning, integrating heterogeneous data and local knowledge from different studies into computational descriptions of organisms that can be tested experimentally.

1.7 What AI can really mean for biology

In my view, AI's most promising contribution to biology is its potential to bring dispersed observations together. By learning complex relationships from large datasets, it could help researchers identify the most promising hypotheses to test and predict the system-wide changes that interventions may cause. In this approach, experiments generate observations and test predictions, while AI helps organize data, build models and make predictions. Results from new experiments then inform further revisions to the models (*King et al., 2009*). This cycle could help researchers make better use of limited experimental resources to understand complex biological processes.

However, the ability to process high-dimensional data does not mean that AI can reliably predict living systems. A more fundamental question is whether biological systems, despite their complexity, exhibit patterns of change that can be learned, modeled and tested. At what scales and under what conditions do these patterns hold? The next chapter examines these questions and the biological assumptions underlying the modeling of living systems.

Chapter 2 | Why biological systems remain modelable despite their complexity

Biological systems contain vast numbers of interacting molecular components. These components interact and change across multiple spatial and temporal scales. Building an effective model begins with defining what we want to predict, over what time frame and with what degree of accuracy. For a given research task, a model can capture the relevant structures and dynamic relationships without reconstructing every molecular event.

The repeatability and stability of processes such as development, physiological regulation and tissue repair suggest that there are underlying patterns we can model. This chapter explores these patterns and what they mean for model design, focusing on four aspects: dynamical constraints, multiscale organization, history dependence and macroscopic observations.

The history of science has repeatedly shown that a complete microscopic description is not a prerequisite for useful prediction. Fluid mechanics does not require simulation of every water molecule, statistical mechanics does not track every atom in a gas, and weather forecasting does not follow every air molecule. These models succeed because complex systems can exhibit higher-level regularities that remain predictable despite microscopic uncertainty.

We believe the same principle applies to biological systems. Their complexity is immense, but their behavior is not arbitrary. Despite billions of years of evolution and countless environmental disturbances, developmental programming, physiological regulation and tissue repair after injury continue to follow reproducible patterns. These processes reflect stable constraints, recurring patterns and structured dynamics. A human embryo develops into a human, rather than a random collection of tissues and organs. A regenerating liver remains a liver rather than becoming a heart or another organ. Stem cells differentiate into a limited range of cell types along pathways illustrated by Waddington's landscape. These observations indicate that biological dynamics are strongly constrained. Complexity does not make biological systems impossible to model; it changes the kinds of models we need. The constraints and patterns described above provide a biological basis for developing digital organism models tailored to specific tasks.

The following four working hypotheses guide our approach to modeling. They draw on established biological observations and extend them into assumptions about model representations, prediction scales and the dynamics of biological systems. Existing research provides evidence for the underlying biological regularities. Whether our modeling assumptions hold, which problems they apply to and whether they improve predictions still need to be tested through data analysis, model comparison and independent experiments.

2.1 Hypothesis 1: Biological systems exhibit constrained dynamics.

Living systems exhibit randomness at the microscopic level, but their macroscopic behavior is constrained by genetic, developmental and physiological regulatory mechanisms. Genetic and developmental mechanisms constrain which structures and behaviors can emerge. Physiological feedback regulates an organism's responses to internal and external changes, while natural selection can preserve heritable traits that favor survival and reproduction across generations. Randomness contributes to biological processes, but it cannot by itself explain their reproducibility, directed responses or continuity across generations.

Biological systems may exhibit chaotic dynamics at certain scales or in particular processes, but this does not mean that the system as a whole is dominated by chaos. Structural constraints and feedback mechanisms shape development, physiological regulation and behavior, allowing certain macroscopic outcomes to recur even in the presence of perturbations. For example, a study of Drosophila embryos shows substantial variation between embryos in the spatial concentration gradient of the maternal morphogen Bicoid. Yet noise filtering by the gene regulatory network keeps the expression boundary of the downstream gene *hunchback* highly consistent. This boundary also remains stable as temperature changes (*Houchmandzadeh et al., 2002*). A study of BMP gradients likewise shows that specific molecular transport mechanisms buffer the effects of changes in gene dosage on spatial patterning (*Eldar et al., 2002*). Together, these findings provide concrete evidence for studying biological relationships that remain reproducible under perturbation.

Strong biological constraints are also evident at several other levels:

- The central dogma is broadly applicable across species (*Crick, 1970*).
- Mendel's laws of inheritance apply broadly to nuclear genes in sexually reproducing eukaryotes.
- Embryonic development within a species shows repeatable timing and morphology. In mice, key events such as gastrulation, somitogenesis, and limb-bud formation occur in a characteristic sequence, producing developmental stages that can be recognized across individuals (*Theiler, 1989*). This repeatability reflects the joint constraints of genetic programs, cell-cell signaling, and tissue interactions.
- Tissue regeneration also reflects internal biological constraints. Compensatory growth after partial hepatectomy in humans (*Nagasue et al., 1987*) and structural regeneration after limb amputation in salamanders (*Kragl et al., 2009*) demonstrate orderly regulation of tissue growth and form.
- In *Stegodyphus lineatus*, for example, young spiders receive regurgitated food from their mother and then consume her through matriphagy (*Salomon et al., 2015*). They later disperse and begin building webs and capturing prey independently (*Pasquet et al., 1999*). These complex, species-typical behaviors emerge during development and illustrate order and constraint at the behavioral level.
- Cell differentiation provides another example of constrained biological change. Waddington's landscape represents alternative differentiation paths as branching valleys (*Waddington, 1957*) , along which cells progressively acquire the structures and functions of specific cell types, such as hepatocytes, muscle cells or neurons. The regulatory constraints underlying differentiation coordinate changes in gene expression and help make the establishment and maintenance of cell fates reproducible under similar conditions.

The modeling hypothesis in this section is that, for a clearly defined prediction task, some regulatory constraints can be represented as dynamic relationships that can be learned and tested.

2.2 Hypothesis 2: Biological systems are hierarchical, multi-scale, self-organizing complex systems.

Biological organization spans molecules, cells, tissues, organs and whole organisms. These levels are linked through the exchange of substances, signaling and mechanical interactions. Local changes can therefore affect tissue and organ function, while conditions across the organism can, in turn, influence cellular and molecular processes. These connections are mediated by information flow across biological levels and scales, coupling the dynamics of biological processes across scales.

- α/β-tubulin heterodimers self-assemble into microtubules through longitudinal and lateral interactions (*Weisenberg, 1972; Nogales et al., 1999*). Microtubule dynamics, together with the activities of molecular motors and other microtubule-associated proteins, contribute to spindle organization and accurate chromosome segregation (*Mitchison & Kirschner, 1984; Kline-Smith et al., 2004*), thereby linking molecular interactions to cell division.

- Dopamine released by dopaminergic neurons acts as a neurotransmitter, changing the membrane potential and synaptic plasticity of recipient neurons and influencing reward-related learning. This connects molecular signals, changes in neural circuits, and individual behavior (*Steinberg et al., 2013*).

- Under specific culture conditions, intestinal stem cells and their descendants can proliferate, differentiate, and interact to form organoids with some features of intestinal tissue. This illustrates the link between cellular interactions and tissue structure formation (*Serra et al., 2019*).

- In mammals, perceived threats can elicit rapid increases in circulating epinephrine and cardiovascular activity (*Breznitz et al., 1998*). Central autonomic pathways regulate sympathetic input to the adrenal medulla, where neural stimulation promotes catecholamine secretion by chromaffin cells (*Strack et al., 1989; Stroth et al., 2013*). Circulating epinephrine acts through adrenergic receptors and contributes to the regulation of cardiac activity, hepatic glucose output, and skeletal muscle energy metabolism (*Motomura et al., 1990; Dufour et al., 2009; Watt et al., 2001*). Together, these mechanisms illustrate how neural activity, cellular secretion, and molecular signaling coordinate physiological functions across scales and organs.

- When guard bees detect a threat, their sting-associated glands release alarm pheromones, such as isoamyl acetate, that can recruit other bees to defense. Chemical communication between individuals shapes the colony response. Information is thus transmitted across scales, from molecules to individual behavior and the superorganism (*Nouvian et al., 2015*).

Together, these examples show how interactions within and between biological levels can give rise to coordinated functions at larger scales. Such coordination does not require a central controller to direct every molecular event. Instead, it can arise from local interactions among biological components, shaped by processes such as diffusion, active transport, protein turnover and ligand-receptor binding, together with feedback regulation. For modeling, the key question is which details must be represented explicitly and which can be summarized at a higher level. **Not every biological scale needs to be modeled at the same resolution.** For a given task, tissue-level variables can summarize some microscopic processes and limit model complexity. Just as aerospace engineering does not need to simulate every atom in a wing to make flight possible, a

virtual organism may produce useful, interpretable predictions without simulating every molecular collision.

The modeling hypothesis in this section is that, for a clearly defined prediction task, higher-level state variables can adequately summarize some microscopic processes. Models operating at different resolutions can then be linked through key interactions across scales to predict the relevant system responses. Multiscale observations and perturbation experiments are needed to determine which processes can be simplified and which interactions must be preserved.

2.3 Hypothesis 3: Biological systems exhibit memory and history-dependent dynamics across multiple temporal scales.

An organism's present state reflects not only current inputs but also the lasting effects of earlier states, external stimuli and past responses:

- In some neural circuits, brief high-frequency stimulation induces a persistent increase in synaptic transmission, known as long-term potentiation (LTP). As an experimental model for studying memory, LTP illustrates how past activity can have lasting effects on subsequent synaptic responses (*Bliss & Lømo, 1973; Whitlock et al., 2006*).
- DNA methylation, histone modifications and chromatin accessibility established during development, tissue repair or environmental stimulation can persist after the stimulus has ended and be maintained or re-established through cell division. By influencing gene expression in descendant cells, these changes provide a molecular mechanism through which past experience can have lasting effects on cell function. For example, in the CRISPRoff study, transient epigenome editing induced gene silencing that persisted through cell division, demonstrating that a brief intervention can have lasting effects (*Nuñez et al., 2021*).
- During mouse sensory neuron development, cells acquire distinct subtypes through a series of differentiation branches, with intermediate cells at each branch point retaining the potential to follow either of two fates. As differentiation proceeds, genes associated with one fate become coordinately expressed while those associated with the alternative fate are suppressed. Because these changes occur before cells acquire their final subtype identities, they suggest that early gene expression patterns contain information about later differentiation (*Faure et al., 2020*).
- Key tumor-driver mutations, such as KRAS G12D and TP53 loss-of-function mutations, make cells more likely to enter trajectories associated with abnormal proliferation and tumor development (*Tuveson et al., 2004; Jackson et al., 2005*).
- After infection with pathogens such as variola virus, measles virus, varicella-zoster virus, or poliovirus, memory T and B cells can persist for decades. On encountering the same pathogen again, the organism can mount a faster or stronger specific secondary immune response (*Hammarlund et al., 2010; Amanna et al., 2007; Crotty et al., 2003*).
- Age itself represents accumulated historical information, providing a temporal reference for development and aging. The lasting effects of development, environmental exposure, injury and repair can remain reflected in molecular features, tissue structure and organ function (*Horvath, 2013; Belsky et al., 2015; Joehanes et al., 2016; Naik et al., 2017*).

Models of virtual organisms therefore need to account for both the present biological state and the effects of past development, environmental exposures and interventions. By learning how past events leave lasting effects on biological states and how those states evolve over different timescales, a model may predict how tissues, organs and whole organisms are likely to change without reconstructing every underlying molecular event.

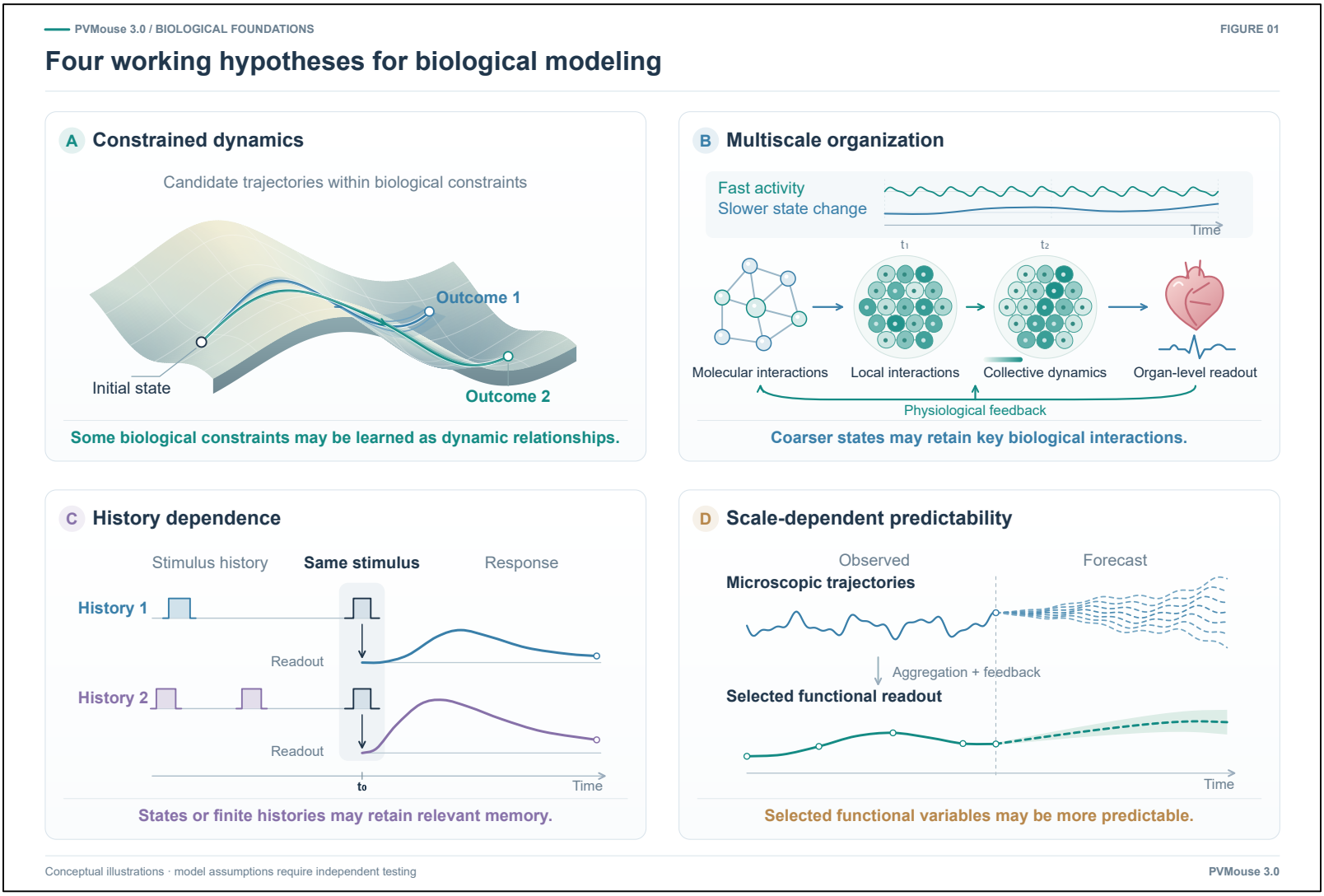
The modeling hypothesis in this section is that some historical influences on future responses can be captured by current state variables or a limited record of past states and events. When current observations do not capture these effects, adding records of development, exposure, and previous interventions may improve prediction. What historical information to retain, and how far back it should extend, depend on the biological process and prediction target.

2.4 Hypothesis 4: Macroscopic physiological states may be more predictable than the microscopic trajectories that generate them.

For a defined research task, macroscopic variables such as organ function, metabolic measures or disease progression may be more suitable prediction targets than individual molecular events. Feedback regulation, redundancy and the combined action of multiple components can buffer microscopic fluctuations, maintaining variables such as blood glucose, body temperature, organ function and anatomical structure within certain ranges. Such stability depends on the conditions and does not, by itself, guarantee reliable prediction under new conditions.

Under some conditions, aggregation over space and time can reduce local fluctuations, while feedback regulation further limits deviations. These mechanisms can make system-level behavior more stable than the activity of individual components, although their effects depend on how those components interact and are regulated. Just as temperature and pressure can be predictable despite uncertainty about the motion of individual molecules, some macroscopic physiological variables may be more stable and predictable than the molecular processes underlying them. In Drosophila embryonic cells, for example, transcription is highly stochastic and occurs in bursts. But the accumulation and local diffusion of mRNA from some patterning genes can average out these fluctuations over space and time (*Little et al., 2013*). This greater predictability does not imply that biological dynamics can be described in fully deterministic terms.

The modeling hypothesis in this section is that, for a clearly defined prediction task, some macroscopic physiological variables exhibit reproducible patterns of change. A model that captures the relevant microscopic mechanisms and system-level constraints may therefore predict how these variables change and respond to interventions without reconstructing individual molecular trajectories.



Four working hypotheses for biological modeling. (A) Biological constraints may be learned as dynamic relationships that restrict candidate trajectories and outcomes. (B) Local molecular and cellular interactions can support collective dynamics and organ-level function, with physiological feedback across scales. Fast activity and slower state change are illustrated over a common time window; the cell diagrams and activity shading indicate local interactions and coordination. Coarser states may retain key biological interactions. (C) Different stimulus histories can lead to different responses to the same subsequent stimulus, motivating states or finite histories that retain relevant memory. (D) Selected functional variables may be more predictable than microscopic trajectories through aggregation and feedback. These are working hypotheses illustrated conceptually, not validated model capabilities. Their applicability depends on the task, biological context, observation scale and prediction horizon.

The biological observations discussed in this chapter support four working modeling hypotheses. Together, these hypotheses provide a starting point for investigating which changes in complex living systems can be reliably predicted.

Chapter 3 | Why biology needs a state space

Chapter 2 explored why, from a biological perspective, complex living systems may still be modeled effectively. Applying these insights to predictions of disease progression or responses to interventions requires us to bring dispersed experimental observations together. Molecular, cellular, tissue and physiological measurements capture different aspects of a living system, while measurements taken at different time points offer clues to how it changes. This chapter asks how these observations can be organized to describe the system's current state and help predict how it will change.

We use a **biological state space** as the framework for organizing experimental observations across biological levels and time points. For a defined biological system and prediction task, a set of variables describes the system's state, and changes over time and under interventions are represented as trajectories. Within this framework, AI helps us learn representations suited to the task from high-dimensional, heterogeneous observations, and use data collected over time and under perturbations to learn how biological states change. This chapter introduces the state-space concept, explains its biological meaning and computational foundations, and then sets out how biological world models and PVMouse 3.0 fit within this framework.

3.1 From biological observations to biological states

Single-cell transcriptomes describe molecular features of cells, while spatial omics reveals how these features are distributed within tissues. Physiological, biochemical, pathological and functional measurements describe organs and whole organisms. These observations capture different aspects of living systems. To predict disease progression or responses to interventions, however, we need to understand what they reveal collectively about a system's current state. Which differences matter for subsequent responses? Which observed differences mainly reflect the measurement process? And what relevant information remains unobserved?

The relationship between the genome and cell function illustrates why we need the concept of a biological state. Coding and regulatory sequences influence biological activity, and genetic variation affects development, disease sensitivity and intervention responses. Yet most somatic cells within an individual share essentially the same nuclear genome while forming distinct cell types, including hepatocytes, neurons and muscle cells. Their functions depend jointly on gene regulation, protein composition and activity, metabolism and the surrounding tissue microenvironment. Even for the same immune cell type, cells that are resting, activated or persistently stimulated may differ in their capacity to respond. A similar distinction applies at the organism level. An individual's genetic background remains essentially unchanged with age, while cell composition, tissue structure, metabolic regulation and organ function change over time. Likewise, individuals with the same disease diagnosis may differ in their disease course and treatment history. Genotype, cell type and diagnosis therefore provide essential context, but prediction also requires a description of the system's current state within that context.

For a specified biological system, scale and task, we describe its current biological state using a set of variables that retain information relevant to subsequent changes. These variables may describe gene expression, protein activity, metabolism, cell composition or tissue function. They may also include latent features learned from multiple types of observation. When

biological processes depend on their history, molecular changes left by earlier stimuli or records of past treatments may provide additional information needed for prediction.

Building such a representation involves three related challenges.

First, preserving differences relevant to the prediction task in high-dimensional observations. Some genes are co-regulated, so a suitable representation of transcriptomic data can capture their coordinated changes while reducing redundancy. However, compression can also discard signals that are small in magnitude but important for subsequent responses. Reconstructing the overall expression profile should therefore not be the only objective. The representation must also preserve information relevant to functional differences and intervention outcomes.

Second, distinguishing biological differences from differences introduced by measurement. Sample processing, the anatomical extent of sampling and the measurement platform can all affect the observations. Even studies of the same biological process over time may organize sample collection and processing differently. One study may begin all experiments on the same day, store samples collected at earlier time points and prepare all sequencing libraries together at the end (*Lawrence et al., 2022*). Another may start experiments on different dates so that fresh samples at different stages of the response can all be collected and processed for library preparation on the same day (*McFarland et al., 2020; Ravindra et al., 2021; Bak et al., 2023*). Both designs allow observations across time, but interpreting expression differences also requires accounting for how samples were stored and processed.

Batch-correction methods such as mutual nearest neighbors (MNN) must therefore be applied in the context of the experimental design (*Haghverdi et al., 2018*). When time points are confounded with differences in sample processing, expression similarity alone cannot determine which differences should be removed. Correction may inadvertently eliminate genuine biological signals that change over time. Preserving this experimental context is essential for interpreting differences between observations when modeling biological states.

Metadata are therefore an integral part of state modeling. Age, sex, genetic background, tissue region and treatment history establish the biological context of an observation, while records of the platform and sample processing help explain differences introduced by measurement. Our project's analysis of GEO and SRA data included in ARCHS4 found that, among more than 700,000 mouse tissue bulk RNA-seq samples, fewer than 4% provided all four basic metadata fields: age, sex, organ and strain. This finding highlights a substantial gap between the scale of public data and the amount that can be used directly to model biological states and responses under specified conditions.

Finally, linking observations across scales through biological relationships and an understanding of what each measurement captures. Single-cell gene expression, bulk tissue expression and tissue function are interconnected. Bulk expression is shaped by both the tissue's cell-type composition and the RNA contribution of each cell type (*Wang et al., 2019*), while functional measurements reflect the combined effects of multiple biological processes. The model must use these relationships so that different observations jointly constrain its estimates of biological state. Simply providing the measurements as separate inputs does not establish those connections. The value of a shared representation is that previously dispersed measurements can complement one another in addressing the same prediction task.

AlphaFold2 offers an example of how representations can be designed for a specific task. It brings together sequence and evolutionary information, structural data and geometric constraints in a computational representation suited to protein structure prediction, substantially improving predictive performance (*Jumper et al., 2021*). This example illustrates why organizing information and introducing task-relevant constraints are important when using large-scale biological observations to make predictions.

3.2 Biological state space: connecting different observations with dynamic prediction

Section 3.1 introduced biological state and explained why genomic information alone does not determine a system's current state. Building on this distinction and the four modeling hypotheses in Chapter 2, we now need a conceptual and mathematical framework for describing biological change and predicting intervention outcomes. A biological state space provides such a framework.

This white paper defines a biological state space as the set of biologically possible or reachable combinations of state variables for a specified biological system, scale and research task. At a given time, the system's state corresponds to a point in this space. As the state changes—for example, in response to environmental changes or drug treatments—that point traces a trajectory. Because experimental observations usually capture only part of the relevant information, the estimated state may instead be expressed as a probability distribution over possible states.

The choice of state variables depends on the research question. A study of liver responses to an intervention, for example, may need to describe lipid metabolism, inflammation, tissue injury, cell-type composition and liver function. Samples with the same disease diagnosis can differ along these dimensions and therefore occupy different positions in state space. Changes in these variables during disease progression or following intervention define trajectories through that space.

The prediction task also determines which observations can be used as inputs. Estimating an animal's unmeasured readouts from its measured tissue expression involves inferring its state from partial observations. Predicting a tissue's response to an intervention, by contrast, requires using the specified biological background and intervention conditions without access to the target response measurements. Distinguishing these tasks clarifies which state variables and inputs are needed and what predictive capability an evaluation actually demonstrates.

State, **observation** and **representation** play different roles. A state describes the system's biological condition; observations are experimental measurements; and a representation consists of the variables and coordinates used to encode state information. Transcriptomic, imaging and biochemical measurements each capture aspects of the system and are affected by sampling and measurement error. An observation model links a biological state to the readouts it could produce under specified measurement conditions, allowing predicted readouts to be compared with actual observations to identify states consistent with the evidence. Different experiments can therefore constrain different aspects of the same biological state through their respective observation models.

In the liver, for example, a change in gene expression may reflect altered expression within hepatocytes, a shift in the proportion of immune cells, or both. A single-cell transcriptomic reference can help identify possible contributions from different cell populations, spatial observations constrain where changes occur, and pathological and functional measurements

indicate whether these changes accompany tissue injury or altered function. Together, these observations can help distinguish changes in cell-type composition, gene expression and function that bulk measurements alone cannot uniquely resolve.

State representations can combine interpretable variables, such as cell proportions, biochemical measurements and the expression of functionally related genes, with learned latent features that capture patterns difficult to specify explicitly in advance. Variables that consistently change together may be summarized using fewer coordinates, simplifying computation. The resulting representation must retain the information needed to predict the outcomes of interest, and its predictive value must be tested in new samples and conditions.

A multiscale representation need not describe every process at the same resolution. The cell layer can retain regulatory and metabolic features, the tissue layer can describe regional distributions and local cell-type composition, and the organ and organism layers can summarize functionally relevant properties. Different levels are linked by aggregating lower-level information and by biological processes such as material exchange and signaling. The shared state space connects descriptions across these levels, allowing local observations to inform estimates of the overall biological state. Whole-body physiological conditions can, in turn, help explain local responses.

The role of genomic information depends on the biological system and timescale. For short-term interventions in mice with a specified genetic background, strain and genotype can usually be treated as relatively stable background conditions. During tumor development and evolution, however, somatic variant profiles and the proportions of cells carrying particular variants can change (*Navin et al., 2011*). The model must capture these changes and their relationships to cell function, tumor progression and treatment response. Background conditions, current state, and external interventions should have distinguishable roles in the model so that the sources of predicted changes can be understood.

Within a defined task, the shared state space brings together information about biological states from different types of observations. It provides a common basis for comparing samples and describing changes over time as trajectories. Distances between state representations can summarize differences, but their biological meaning depends on the variables, scales and distance measures used. Whether movement toward healthy samples in latent space corresponds to functional improvement must be tested against relevant physiological and pathological endpoints. Once supported by such evidence, comparisons between states can help evaluate interventions. Predicting intervention outcomes, however, also requires understanding how a system moves from one state to another over time. These state transitions are the focus of the next section.

3.3 Modeling diseases and intervention responses through state transitions

Sections 3.1 and 3.2 introduced biological states and state space. In classical mechanics, a description of position and momentum is followed by the study of how they change over time. Biological modeling similarly asks how a system moves between states, what maintains a particular state and how its trajectory changes when conditions change. We summarize this perspective as **Biology is the science of state transitions.** From this perspective, disease progression, development, aging and responses to interventions can all be studied as state transitions.

Disease progression illustrates why a diagnosis alone cannot describe a biological state or its likely trajectory. In metabolic dysfunction-associated steatotic liver disease (MASLD), including its steatohepatitis form, metabolic dysfunction-associated steatohepatitis (MASH), disrupted lipid metabolism, insulin resistance, cellular stress and immune responses contribute to pathology. Some individuals develop fibrosis, while others remain stable for long periods (*Sanyal et al., 2021*) or improve following intervention (*Harrison et al., 2024*). Because lipid accumulation, inflammation, tissue remodeling and functional changes need not progress together, a single diagnosis cannot describe the extent of each process.

A biological state space can make this variation within a single diagnosis explicit by representing the underlying processes through molecular, cellular and tissue variables. Two groups with similar lipid accumulation may respond differently to the same intervention because they differ in inflammation, fibrosis or functional reserve. Observations over time can also distinguish apparently similar states that are moving toward progression or improvement. The current state describes the system at a particular time, while its trajectory shows how that state develops and how earlier changes may influence later responses.

Because intervention responses depend on both the initial state and the changes that led to it, the same target may have different effects at different stages of disease. For example, a state dominated by lipid accumulation presents different opportunities for treatment and recovery from one involving extensive tissue remodeling. A single target may still offer an effective point of intervention, but the outcome depends on how the effects of modulating it propagate through the system and are shaped by feedback, redundancy and compensatory responses. These effects must ultimately be assessed through functional outcomes. State-transition models therefore link target selection to the initial biological state, intervention timing and subsequent outcomes.

Cellular responses provide another example. Shared markers do not fully explain how cells in different states will respond to treatment. In cancer immunotherapy, antigen stimulation can lead naive $CD8^+$ T cells along different differentiation trajectories associated with effector function, memory or exhaustion. Under persistent antigen stimulation, some tumor-reactive $CD8^+$ T cells acquire a progenitor-exhausted state and can differentiate through intermediate states into terminally exhausted cells (*Beltra et al., 2020*).

Tumor-infiltrating $CD8^+$ T cells can differ in differentiation potential and treatment response even when they all express PD-1. Progenitor-exhausted T cells can maintain their population and produce more differentiated descendants, contributing to expansion and tumor control following PD-1 blockade (*Siddiqui et al., 2019*). Terminally exhausted cells differ in their capacity to persist and respond to treatment (*Miller et al., 2019*). A model must therefore represent the states and abundance of these populations, together with transitions between states. Their composition before intervention can inform predictions of which populations will expand or differentiate after treatment and how those changes relate to tumor control. Measurements of cell states, population sizes and tumor outcomes at different time points can then test these predictions.

Interventions can also be designed to guide state transitions. Small molecules, antibodies and other perturbations seek to alter disease-related molecular networks and cellular activity to reduce pathology and restore function. Induced pluripotent stem cell (iPSC) reprogramming provides another example. Following the introduction of Yamanaka factors under suitable culture conditions, fibroblasts can progressively establish regulatory programs associated with pluripotency (*Takahashi & Yamanaka, 2006*). The starting fibroblasts and resulting pluripotent

stem cells define the initial and final states of successful reprogramming. Measurements at multiple time points help distinguish trajectories leading to successful reprogramming from those associated with apoptosis (*Marión et al., 2009*) or partial reprogramming (*Mikkelsen et al., 2008*).

Development and aging can likewise be described through state transitions. Development involves cells following differentiation trajectories toward specific identities, while aging encompasses changes across molecular, cellular, tissue and organism levels. These examples show why a measurement at one time point is only a starting point for understanding biological processes. Following changes over time can reveal intermediate states, distinguish trajectories and identify opportunities to alter later outcomes through intervention.

For drug development, this perspective raises three related questions: which initial disease states are most likely to benefit from an intervention; how responses depend on intervention strength, timing and sequence; and whether molecular and cellular changes lead to measurable functional improvement. Drug combinations and sequential treatments can be represented as inputs that vary over time. Chapter 7 discusses how specific regimens can be compared and selected.

Intervention sequence makes the importance of state particularly clear. An earlier treatment may alter cell-type composition, feedback activity or the tissue microenvironment, changing the conditions under which a later treatment acts. Administering A before B may therefore produce a different outcome from administering both simultaneously (*Lee et al., 2012*). The model needs to describe how the state produced by one intervention influences the response to the next. Dosing schedules, treatment duration and prior interventions thus become part of the same dynamic prediction task.

Evaluation should use both the untreated disease course and a healthy physiological reference. Comparison with the untreated course helps estimate what the intervention changes, while the healthy reference helps interpret the physiological significance of those changes. Gene expression profiles and distances between state representations provide information about biological changes, but their consequences must be assessed using disease-specific functional and pathological endpoints. Improvement in some measurements but not others may reflect differences in the timing of recovery or indicate an important process missing from the representation.

State-transition modeling brings cell differentiation, disease progression and intervention responses into a shared framework. It offers a basis for predicting intervention outcomes from observed biological states, but poses substantial computational challenges. These include high-dimensional states, nonlinear relationships, dependence on context and history, and interactions between multiscale feedback and biological constraints. AI methods, particularly foundation models, offer tools for learning state representations from large biological datasets, integrating different types of observations and predicting changes under specified interventions. The next section examines how these methods can learn biological states and transitions, and how time-series data, perturbation experiments and independent validation can test the biological value of their predictions.

3.4 Learning biological state representations and dynamics with AI

The previous section framed development, disease progression and intervention responses as state-transition problems. Under the working hypotheses introduced in Chapter 2, living systems can be viewed as high-dimensional dynamical systems subject to biological constraints. Predicting how

their states change requires learning representations of the current state from observations collected under different conditions and relating those representations to subsequent outcomes. AI provides computational methods for learning both state representations and predictive relationships.

Recent advances illustrate several contributions of AI to biological modeling. AlphaFold 3 has advanced the prediction of biomolecular structures and interactions (*Abramson et al., 2024*), while Evo 1 and Evo 2 have been used to design bacteriophage genomes (*King et al., 2026*). Geneformer, scFoundation and scGPT learn representations from large-scale single-cell datasets for use in downstream analyses, while STATE and ProteinTalks (*Sun et al., 2026*) explore the prediction of cellular responses to specific perturbations. Researchers have also proposed integrating models across scales to build digital organisms (*Song et al., 2026*). Although these studies address different problems, they offer methods for extracting useful representations and predictive relationships from complex biological data, providing starting points for modeling biological states and their changes.

AI can therefore support two complementary tasks: learning reusable state representations from observations, and learning how states change from temporal and perturbation data. Foundation models provide an initial basis for representation learning. The specific research task then determines what additional information the representation must preserve, which relationships the model must learn and how its predictions should be evaluated.

3.4.1 Learning Representations from Observations

AI models can learn representations of biological samples by training on prediction tasks using large datasets. In masked gene expression reconstruction, for example, some expression values in a cell are hidden, and the model learns to predict them from the remaining information. Prediction errors are used to adjust its parameters. If a group of genes consistently changes together during cell activation, the model may learn to infer the expression of some genes from that of others. Training across many cells and conditions helps the model capture expression relationships that support prediction.

To perform these tasks, the model transforms gene expression data into an internal numerical representation. Each cell is represented by a set of coordinates in a latent space, where cells can be compared according to the features learned by the model. Individual coordinates do not necessarily correspond to distinct biological processes. Patterns across multiple coordinates, however, may capture information like cell identity, regulatory activity and functional state.

Expression reconstruction provides a starting point for representation learning, but the research task determines what else a representation must preserve. For example, features associated with lineage may help distinguish immune cell types, while distinguishing resting and activated cells within the same type requires information about their activity. A representation developed for one task may therefore need to be adapted for another. For biological state modeling, it must preserve information needed to predict the target measurements and subsequent responses of interest.

Functional observations can help the model learn representations that retain information relevant to functional outcomes. After self-supervised training, such as masked expression reconstruction, paired expression and functional measurements can be used to train the model for both expression reconstruction and functional prediction. For example, two immune cell populations may have similar overall expression profiles but differ in measured function. If the

input data contain signals associated with that difference, training on functional outcomes can encourage the model to preserve those signals, even if they contribute little to overall expression reconstruction. The representation can thereby preserve information relevant to functional differences as well as relationships among expression measurements.

Whether a learned representation is useful for describing biological states must be tested with independent data. Sequencing depth, experimental batch and other measurement-related factors can also influence the representation. Evaluation should examine whether comparable biological features are represented consistently across studies, whether functional differences within the same cell type remain distinguishable, and whether the representation supports prediction of target measurements in samples not used for training. The results can guide revisions to the training tasks and identify gaps that require additional observations. If supported by these evaluations, the learned latent space can serve as a coordinate system for representing biological states within the defined task.

3.4.2 Evidence from Existing Studies

Single-cell foundation models have demonstrated a workflow of learning representations first and then adapting them to specific tasks. After pretraining on large single-cell transcriptomic datasets, Geneformer (*Theodoris et al., 2023*) was adapted to predict gene dosage sensitivity and chromatin dynamics and to identify candidate therapeutic targets in cardiomyopathy research. scGPT (*Cui et al., 2024*) uses pretrained representations for cell type annotation, integration across experimental batches and perturbation-response prediction. scFoundation (*Hao et al., 2024*) has demonstrated applications of learned expression representations to drug-response prediction and the prediction of genetic perturbation effects.

These studies provide a basis for reusing learned representations rather than training a new model from scratch for every task. Existing foundation models can therefore serve as starting points for biological state representation. Their suitability and practical benefits must nevertheless be evaluated for the target species, cell types and tissues, taking into account the genes included in the input and the data available for the target task.

An independent comparison involving single- and double-gene perturbation prediction illustrates why this task-specific evaluation matters: foundation models and other deep learning methods did not outperform simple baselines on the tasks tested (*Ahlmann-Eltze et al., 2025*). Good performance in expression reconstruction or cell classification does not, by itself, demonstrate the ability to predict intervention responses. Moving from describing current states to predicting state changes requires learning the relevant relationships from temporal and perturbation data and evaluating the resulting predictions against simple baselines.

3.4.3 Learning State Changes from Temporal and Perturbation Data

State estimation describes a system's current condition, while intervention prediction asks how the system will respond to a specified intervention. A model that distinguishes inflammatory from resting cell states may still be unable to predict whether inhibiting a particular gene will move cells from one state to the other. Recognizing the expression patterns associated with each state does not, by itself, establish which changes would cause a transition between them. Temporal observations and controlled perturbations provide evidence for learning these transitions by linking initial states and intervention conditions to subsequent responses.

Measurements at different time points reveal patterns of change, while controlled perturbations help establish how responses depend on intervention conditions. Returning to the MASH example, tissue and functional measurements collected before and after an intervention can help establish the sequence of gene expression responses, inflammatory changes and functional improvement. Data covering different doses and disease stages at treatment initiation can reveal how responses depend on both the intervention and the initial state. The object of learning thus expands from "What state is this sample in?" to "What changes might occur from this state under the given conditions?"

Computationally, the model starts from the current state and predicts the state at a specified future time, taking into account the biological background, the applied perturbations and the time interval between the current and target time points. The predicted state is then mapped to gene expression and functional readouts that can be compared with experimental observations. During training, discrepancies between predictions and observations are used to update model parameters. Training across biological backgrounds, interventions and time points helps the model learn how these conditions relate to subsequent changes. Depending on the experimental design and available data, predictions may be evaluated against later measurements from the same individual or against population-level state distributions at different times.

The STATE (*Adduri et al., 2025*) model illustrates how cell state representations can support prediction of gene expression following perturbation. Its state embedding model learns representations from large single-cell expression datasets, while its state transition model combines cellular expression information with perturbation conditions to predict responses. STATE predicts gene expression responses across a population of cells, capturing both the average response and variation among cells. This provides an example of how learned representations can be connected to perturbation-response prediction. Predicting how these responses change over time also requires data from different time points and a model that captures those changes.

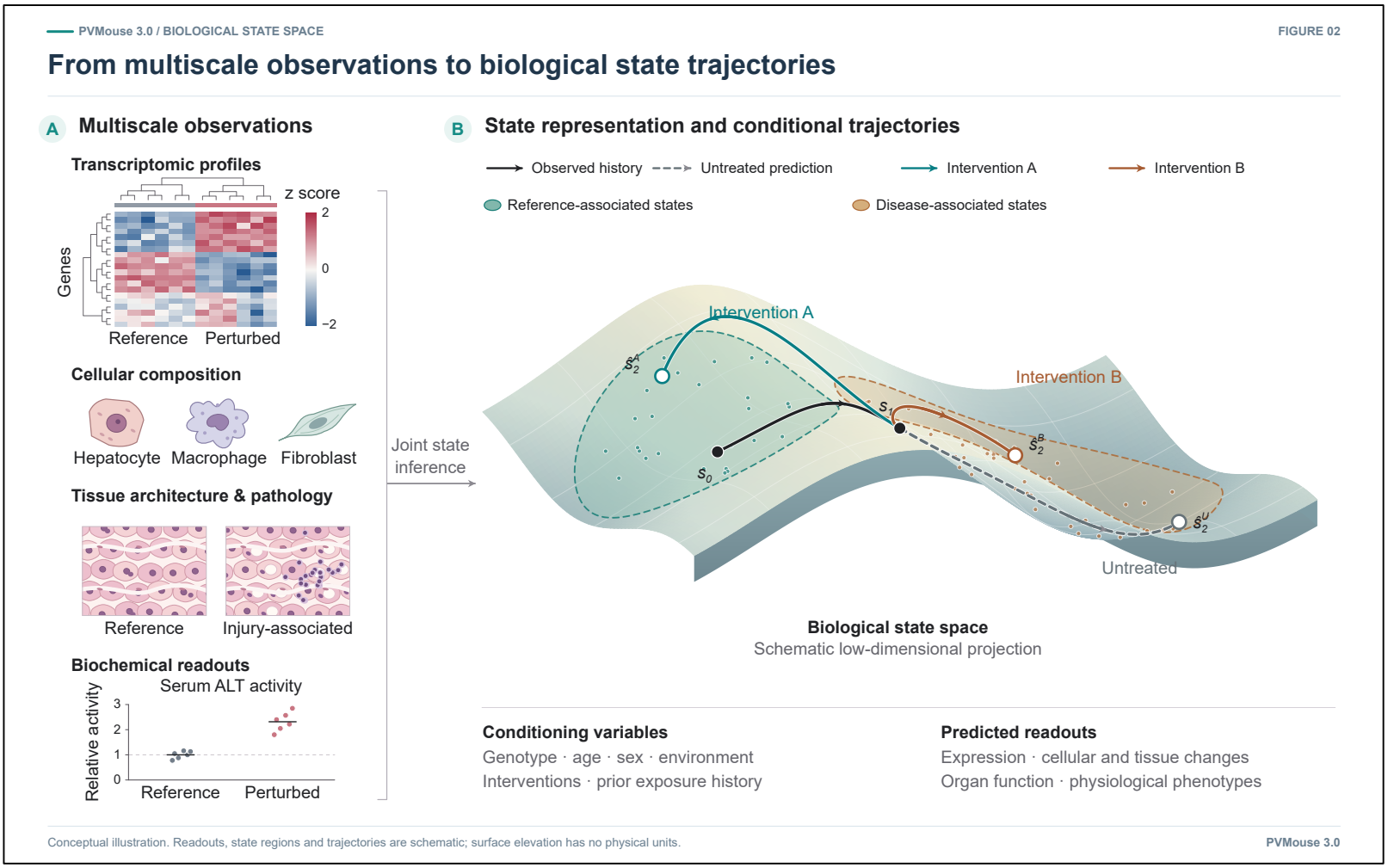
Learning dynamics requires distinguishing population-level changes from individual trajectories. Many omics measurements are destructive, so samples collected at different times often come from different cells or animals. Such data directly describe population state distributions rather than the history of a particular individual (*Schiebinger et al., 2019*). When genetic background, environment and treatment conditions are comparable, explicit assumptions about dynamics can be used to infer candidate trajectories linking these distributions. These trajectories may provide an initial basis for individual-level prediction, but more specific predictions require lineage information, repeated functional measurements or other evidence connecting earlier and later states, followed by independent validation.

The available time points also determine how much temporal detail the data can support. With sparse sampling, modeling may initially focus on changes between specified observations. As measurements cover more stages of a response, the rate, sequence and duration of change can be studied in greater detail. For history-dependent processes, relevant past information must be captured in state variables or supplied through treatment records and repeated measurements. Comparing predictions made with and without historical information can test whether it improves explanation and prediction.

Predicting later outcomes can also help improve state representations. If a model accurately reconstructs current expression but repeatedly fails to predict subsequent function, researchers can investigate whether it omits relevant protein, metabolic or historical information. Researchers can

also check whether the model predicts state changes incorrectly or fails to translate those states into accurate predictions of experimental measurements. Additional observations and revised training tasks can then address the identified gaps. Any resulting improvements must be evaluated at time points and under perturbation conditions not used for training, as well as through independent experiments.

Foundation models provide candidate state representations, temporal and perturbation data constrain models of state change, and independent validation tests predictions and guides revision. Extending this approach across cells, organs and the whole organism requires explicit, testable relationships between levels. These connections provide the basis for the biological world model discussed in the next section.



Multiscale observations and biological state trajectories. (A) Transcriptomic profiles, cellular composition, tissue architecture and pathology, and biochemical readouts provide complementary observations. (B) These observations support joint state inference in a shared biological state space. Observed history, untreated prediction and alternative interventions are distinguished, alongside reference-associated and disease-associated states. Genotype, age, sex, environment, interventions and prior exposure history define the prediction context. Predicted readouts include expression, cellular and tissue changes, organ function and physiological phenotypes. The surface is a

schematic low-dimensional projection with no physical units for its elevation; readouts, state regions and trajectories are illustrative.

3.5 From state model toward a biological world model

This white paper defines a biological world model as a computational system that integrates multiscale observations to estimate current biological states and predict how they may change and what functional consequences may follow. These predictions concern a specified biological system, scale and task, over a defined time horizon and under specified environmental and intervention conditions.

This approach builds on existing proposals. AIDO, introduced by Song and colleagues, uses a modular architecture to connect foundation models across biological scales (*Song et al., 2026*). Noori and colleagues discuss world models that simulate biological trajectories under different interventions across molecular, cellular, tissue and clinical scales (*Noori et al., 2026*).

Beyond representing states and predicting state transitions, a biological world model must specify how models at different scales exchange information and constrain one another. These connections allow the models to work together to predict responses of the system as a whole.

3.5.1 From Local Models to an Interconnected Multiscale System

Living systems span molecules, cells, tissues, organs, and the organism, with continuous information exchange and feedback across these levels. A drug, for example, may bind to molecular targets and alter cellular activity, with subsequent effects on tissue function. Meanwhile, absorption, metabolism and clearance determine drug exposure within tissues and thereby influence cellular responses. Predicting these effects requires linking local processes to overall physiological conditions.

A biological world model connects models at different scales so that they can work together on a shared prediction task. For example, a pharmacokinetic model estimates how tissue drug concentrations change over time. A cell model uses these concentration profiles together with local cell states to predict cellular responses. An organ model then relates cell states and composition to tissue-level measurements and functional outcomes. Changes in tissue states can, in turn, influence cell-level models through changes in metabolites, hormones and other systemic signals. AIDO's modular approach to connecting models across scales provides a reference for building such a system in stages.

Each module can use a representation suited to its scale. The cell layer describes cell states and composition, the organ layer summarizes regional features and functional states, and the organism layer represents relationships among organs. Modules exchange information about task-relevant states and conditions, allowing them to work together without requiring every module to represent all underlying microscopic details. Chapter 5 describes the interfaces and computational methods that support these connections.

3.5.2 Four Core Capabilities of a Biological World Model

To connect the concept of a biological world model with practical research tasks, this project defines four core capabilities:

1. **Current-state estimation:** Infer the current biological state from available molecular, tissue and physiological observations, clearly distinguishing direct measurements from model estimates.
2. **Prediction of state changes:** Predict how biological states and functional outcomes may change over a defined time horizon under specified background and treatment conditions.
3. **Perturbation-response prediction and counterfactual simulation:** Compare possible outcomes of different genetic, drug, or environmental inputs. Interpreting these comparisons as causal effects or individual counterfactuals requires appropriate experimental designs and assumptions that support causal identification.
4. **Uncertainty estimation:** Characterize uncertainty in predictions and identify gaps in supporting evidence to help prioritize additional observations and experiments.

Together, these capabilities support a workflow for answering biological questions. The model estimates an initial state from available observations, predicts subsequent states and measurable outcomes under the conditions being compared, and reports the evidence supporting those predictions. Predictions should reflect both uncertainty in the initial state and limits in the coverage of perturbation data. The model's appropriate uses therefore depend on the specific task and the evidence obtained through validation.

3.5.3 An Example: Drug-Induced Liver Injury (DILI)

Consider a comparison of the liver risks associated with two candidate compounds at specified doses and dosing schedules. DILI may involve reactive metabolites, mitochondrial injury, cholestasis, oxidative stress and immune responses, with tissue expression changes and functional injury unfolding over time (*Mosedale & Watkins, 2017*). A world model can organize this task into a sequence of linked modeling steps:

- Estimate the initial state and exposure: Use the animal's genetic background, age, available measurements of liver function and dosing history to estimate its initial state. Use available data to characterize how drug exposure in the liver changes over time.
- Predict cellular responses: Combine exposure estimates with local cell states to predict changes in gene expression and metabolism, as well as potential injury responses in hepatocytes and other relevant cell populations. Specify when these changes are expected to occur.
- Predict tissue outcomes: Relate cellular responses to their locations within the tissue, cell-type composition and organ function to predict changes in pathological findings and biochemical measurements.
- Compare candidates and plan validation: Compare predicted outcomes across compounds or dosing schedules under matched evaluation conditions, and identify experiments that would distinguish the most consequential competing predictions.

This workflow translates a risk assessment into testable hypotheses with defined initial conditions, predicted time courses and measurable outcomes. For example, if the model predicts that local cellular stress precedes organ dysfunction, researchers can collect tissue gene expression data, measurements of cellular stress and liver function measurements during the relevant time windows to test whether the changes occur in the predicted sequence.

Experimental results can then be used to update state estimates and revise relationships between modules. Errors in predicted exposure, cellular responses or functional outcomes may indicate different needs for additional data. Interpretations should draw on actual observations, how predictions change under different input conditions and existing biological evidence, making the basis of each conclusion traceable. Chapter 6 discusses how this feedback can support continued learning and the selection of subsequent experiments.

3.6 PVMouse 3.0: A Minimum Viable Biological World Model Starting with the Mouse

Development of a biological world model can begin with an operational MVP whose biological scope is clearly defined and whose predictions can be tested experimentally. PVMouse 3.0 uses the mouse to explore how information about genes, cell states, tissue structure, organ function and organism-level phenotypes can be integrated into a computational framework for representing biological states. Its aim is to generate testable predictions of age-related changes and responses to specific interventions.

The mouse provides a practical starting point because of the wealth of existing data and the ability to control experimental conditions. Relatively consistent genetic backgrounds and standardized environments facilitate comparisons across conditions, while multi-tissue sampling, controlled genetic perturbations and associated phenotyping connect local changes to systemic responses. Human studies face more heterogeneous genetic backgrounds, living environments and treatment histories, together with greater restrictions on repeated multi-tissue observations. Mouse experiments therefore offer a tractable basis for testing multiscale modeling methods.

PVMouse 3.0 initially focuses on the brain, heart, liver, skeletal muscle and spleen in C57BL/6J mice, following the principle of "first reconstruct normal states, then simulate deviations." Data spanning ages, sexes and tissues will establish a population-level physiological reference across ages. Single-gene knockouts, paired multi-tissue transcriptomic data and physiological and pathological measurements from the same animals will support studies of systemic responses to genetic perturbations. Prediction tasks involving drugs, diet and other environmental interventions will be added as relevant data and experimental validation become available.

In this design, public single-cell and spatial atlases provide reference information for PVMouse 3.0 at different biological levels. The project's own multi-tissue data preserve the pairing among tissues from the same animal, while functional and pathological readouts connect states with biological outcomes. Together, these observations support a central test: does adding cellular, spatial, or other-organ information improve prediction of the target response compared with using background conditions or expression data from a single tissue alone? The results will help determine which modules should be prioritized and retained.

To build and test this framework, PVMouse 3.0 will draw on public atlases, knowledge bases, existing foundation models and project resources, allowing new effort to focus on key connections across scales and their validation. Clearly defined interfaces between components will support adaptation of existing representations, replacement of individual modules and the gradual addition of tissues and observation types. Early prototypes will establish links between biological states and measurable readouts. Resolution and coverage will then be increased when this leads to better

predictions. Chapters 5 and 8 describe the computational architecture and the data and resources required for development.

The MVP will be evaluated against measurable outcomes from the outset. Predictions of gene expression and functional readouts will be assessed using data not used for training. Follow-up experiments will help distinguish among competing interpretations and guide model revision. Outputs will include data sources, relevant intermediate predictions and uncertainty estimates so that researchers can examine the evidence behind a prediction, identify sources of error and select subsequent experiments. The initial aim of PVMouse 3.0 is to establish a working multiscale modeling approach that can be compared with alternatives and refined over time, bringing observations from separate studies together to support biological prediction.

Developing PVMouse 3.0 requires specifying what biological information to represent at each level and how those levels are connected. Chapter 4 examines these biological relationships, from molecular interactions and cell states to tissue organization and organism-level phenotypes, laying the foundation for the computational architecture.

Chapter 4 | From Gene Networks to Organism-Level States: The Biological Architecture of PVMouse 3.0

The biological architecture of PVMouse 3.0 must define which biological information to represent at each level and how processes at different levels interact. Gene regulation and biomolecular interactions shape cellular activity; cell composition, spatial organization and intercellular communication contribute to organ function; and organs cross-talk through circulatory, metabolic, immune and neuroendocrine pathways. Observations across these levels help establish these relationships and jointly constrain the model's representation of the whole organism.

Starting from a normal physiological reference for C57BL/6J mice, this chapter outlines the biological processes and relationships to be represented at the molecular, cellular, organ and organism levels. It also explains how genetic perturbations and functional measurements help link information across these levels. Chapter 5 describes how this biological framework is translated into computational modules and how those modules are trained and validated.

4.1 Establishing a Normal Physiological Reference for C57BL/6J Mice

PVMouse 3.0 initially focuses on C57BL/6J mice, following the principle of "**first reconstruct normal states, then simulate deviations**". An inbred strain offers a relatively uniform genetic background, and husbandry conditions can be standardized. Existing molecular atlases and genetic tools further support the integration of multiscale observations with controlled experiments. The model will first be built and validated within this defined scope, using normal physiological states as a reference for interpreting changes associated with disease and interventions.

Mice of the same strain are not biologically identical. Age, sex, diet, housing and previous treatments can affect cellular activity and tissue function. The reference must therefore account for these factors rather than describe a single fixed "normal" state. PVMouse 3.0 will combine gene expression, cell-type composition, tissue structure and physiological measurements to estimate each mouse's biological state, while recording its genetic background, experimental conditions and relevant history. For genetically modified mice, the specific genotype, breeding history, and actual genetic background also need to be recorded to assess whether experimental and control groups are comparable.

Biological context must be distinguished from measurement conditions. The organ and anatomical region identify the biological source of an observation, while sampling, tissue processing, assay platform and experimental batch influence how it is measured. Recording this information for each sample helps distinguish biological differences from variation introduced by the measurement process.

The reference must also capture variation across ages. Development, maturation and aging involve different biological processes, and cell-type composition, tissue structure and organ function may change at different rates (*Schaum et al., 2020*). Observations from mice at different ages will be combined to describe tissue and organism state distributions and relationships among age-related changes across tissues. Population reference trajectories will summarize typical patterns of change across ages, rather than directly trace the course of individual animals. In gene knockout experiments, controls matched for age, sex and genetic background will help assess whether observed differences fall outside the range of normal physiological variation.

Building on this reference, PVMouse 3.0 will model state changes associated with genetic perturbations, disease, drug treatment and other interventions, and assess whether these changes are accompanied by altered organ function or organism-level phenotypes. Deviations may reflect changes in the strength or timing of existing processes. They may also involve cell states or tissue structures not represented in the normal reference data, requiring additional observations to extend the model. A deviation does not necessarily indicate damage; its functional significance must be assessed using physiological, biochemical and pathological measurements.

4.2 The Molecular Layer: Gene Networks Provide Biological Constraints and a Framework for Interpreting State Changes

Gene regulation and interactions among gene products provide a molecular basis for understanding cell states and how they change. The same pathway may play different roles across cell types and ages. PVMouse 3.0 will organize existing biomolecular knowledge to guide and constrain how the model learns changes in cell state and to support biological interpretations that can be traced to supporting evidence.

The molecular layer uses genes as its basic nodes and progressively integrates prior knowledge of transcriptional regulation, protein-protein interactions, ligand-receptor relationships, pathways, gene sets and protein complexes. Different biological relationships, such as regulatory interactions, protein-protein interactions and pathway membership, are represented as distinct relation types. Gene sets and pathways are represented as groups of genes, preserving both their membership and the biological processes or functional characteristics they share. Relations also record the strength of published evidence, tissue specificity, and condition dependence.

Molecular priors must be interpreted in the biological context of the target tissue. Database records may come from different species, cell lines or experimental conditions, so evidence for a relationship does not imply that it is equally active in every mouse cell. PVMouse 3.0 will consider the source of the evidence, the expression of the relevant molecules and tissue function when assessing each relationship's relevance to the cell states being modeled. How this evidence is used will be updated as molecular and perturbation data accumulate.

The molecular layer thus serves three purposes: to provide a structured, interpretable molecular framework and explicit constraints for high-dimensional expression changes; to incorporate regulatory network and pathway knowledge into the learning of state transitions; and to maintain an evolving set of candidate mechanisms. The cell layer described next will build on this foundation, incorporating age, lineage and the surrounding tissue microenvironment to characterize molecular network activity and its relationship to cell function.

4.3 The Cell Layer: Age, Lineage, and Microenvironment Jointly Define Cellular States

The cell layer of PVMouse 3.0 describes both cell type and current state. Hepatocytes, macrophages and fibroblasts have relatively stable lineage identities, but their activity and ability to respond to stimuli vary with age, metabolic load, inflammation, hypoxia, tissue repair and the surrounding tissue environment. Macrophages, for example, can exhibit different gene expression profiles and functions during tissue homeostasis, inflammation and repair (*Wynn et al., 2013*). Cell type labels alone therefore cannot capture all the biological variation relevant to prediction.

PVMouse 3.0 will represent these differences using single-cell transcriptomic observations together with strain, genotype, age, sex, organ and anatomical context. Gene networks and pathway knowledge from the molecular layer will help the model derive features related to regulation, metabolism, stress responses and other cellular activities from gene expression patterns. Because these features initially reflect transcriptional changes, their functional significance must be assessed using protein, metabolic or other functional measurements. Records of previous treatments will also be retained where relevant, allowing tests of whether historical information improves state descriptions and response predictions.

Existing single-cell foundation models, including scFoundation, scGPT and Geneformer, provide a starting point for representing cell states. PVMouse 3.0 will fine-tune these models using mouse single-cell data from multiple tissues and ages, with reliable age information and well-documented experimental conditions. The aim is to interpret cell states in the context of the animal's age and sex, as well as the cell's organ and anatomical origin.

For a given age, sex, tissue and set of experimental conditions, the cell layer will provide gene expression features for different cell types, transcriptome-based estimates of relevant pathway activity and descriptions of how these features vary with age. Outputs will include uncertainty estimates, reflecting limited reference data for rare cell types, transitional states and poorly sampled anatomical regions. These cell state representations will provide the basis for constructing virtual organs. The next section examines how the spatial organization of cells and their interactions contribute to organ function.

4.4 The Organ Layer: Organ States Emerge from Cell Composition, Spatial Structure and Intercellular Communication

Organ states are shaped jointly by cell-type composition, cell states, and their spatial organization. Even when two tissue samples have the same cell-type proportions, their cells may be arranged differently and occupy regions with different specializations. Differences in the distribution of blood vessels and nerves, extracellular matrix organization, contacts between neighboring cells and local signaling can also affect tissue function. The organ layer of PVMouse 3.0 therefore needs to place cellular information within specific tissue structures and describe how different regions contribute to organ activity.

Reconstructing a virtual organ begins with reference data on cell composition and cell states. PVMouse 3.0 will combine single-cell or single-nucleus transcriptomics, spatial omics and tissue imaging to estimate cell types, relative abundances and state features within the anatomical region being modeled. These estimates must account for whether cells or nuclei were sampled, as well as effects introduced by tissue dissociation, nuclei isolation and capture (*Denisenko et al., 2020*). Together, these sources inform estimates of cellular composition in the target tissue or organ.

Linking cell composition to tissue expression also requires accounting for each population's contribution to total tissue RNA. Expression features will therefore be aggregated using relative cell abundances, RNA content and measurement conditions, with observation models connecting single-cell or single-nucleus expression to overall tissue or organ expression. Bulk RNA-seq data from the corresponding anatomical region will constrain these estimates of aggregate expression, linking cell states to tissue-level molecular measurements.

Spatial information helps organize cell states into regional and organ-level representations. In the brain, the Allen Institute's spatial transcriptomic and anatomical atlases provide references for anatomical regions, cortical layers, regional gene expression and cell neighborhoods (*Yao et al., 2023*). PVMouse 3.0 will combine these references with observations from the target sample to estimate the spatial distributions of cell populations and relationships between neighboring cells. These estimates will define regional states, which will then be integrated into an organ state within the modeled anatomical scope. Where spatial measurements are available for the target sample, they will directly constrain the reconstruction. Elsewhere, the model will retain multiple plausible spatial arrangements consistent with the reference atlases and other available evidence. Additional spatial measurements can help distinguish among these alternatives. Regional expression, aggregate tissue expression and spatial organization will each be evaluated against observations at the corresponding scale.

The same approach extends to the heart and other organs: single-cell sequencing identifies cell types and states, spatial omics helps map their distribution within tissue structures, and bulk RNA-seq measures overall expression in the sampled tissue. PVMouse 3.0 combines these sources to construct virtual organs and uses measured bulk expression to test whether they capture age-related changes. Spatial omics, pathology and organ function measurements then evaluate the reconstructed tissue structure and its relationship to organ function.

The organ layer will also gradually introduce intercellular communication. Ligand-receptor priors can suggest candidate communication pathways, whose effects depend on colocalization, cell abundance, receptor expression, protein levels, diffusion distances, and physiological conditions (*Dimitrov et al., 2022*). PVMouse 3.0 will include these relationships together with their sources and evidence strength. As spatial multi-omics, perturbation experiments, and functional readouts accumulate, the model can improve estimates of these relationships and assess their causal roles.

The resulting organ representation should capture task-relevant cell-type composition, regional structure, gene expression features and functional measurements, while clearly distinguishing measured quantities from model estimates. For example, tissue expression changes can be linked to local injury or functional measures to form explanations that can be tested further. Such organ states also provide a basis for studying relationships across organs, allowing local responses to be analyzed together with changes in other organs and organism-level phenotypes.

4.5 The Organism Layer: Joint Multi-Organ States and Cross-Organ Relationships

The physiological state of a mouse reflects the coordinated activity of multiple organs. Metabolites, hormones, immune factors and neural signals connect organs, allowing local changes to affect other tissues, while systemic conditions can, in turn, influence local responses. For example, glucose and lipid metabolism in the liver is linked to energy use in muscle and adipose tissue, while the secretory activity and inflammatory state of adipose tissue can also influence liver function (*Reilly et al., 2015*). The organism layer of PVMouse 3.0 will represent these connections to examine how organ states change together and relate to whole-body physiology.

PVMouse 3.0 will characterize gene expression, cell composition and functional features in each organ across ages and experimental conditions, then link these states through circulatory, metabolic, immune and neuroendocrine relationships. Each organ will be represented as a node,

with connections between nodes representing physiological relationships. Information from other organs can then help estimate the state of a target organ. Multi-tissue observations from the same animal and datasets covering multiple ages, such as a multi-tissue aging transcriptomic atlas (*Shavlakadze et al., 2026*), will provide reference data for joint modeling. Together with intervention experiments, these observations will help the model learn relationships between organs and how those relationships vary with biological and experimental conditions.

Multi-tissue observations from the same mouse are an important basis for joint multi-organ states. These samples share the animal's genetic background, age, sex, and treatment history, reducing variation introduced by artificially pairing tissues from different animals. With data across ages and experimental conditions, the model can analyze which organ states tend to occur together and whether these associations change with age or perturbation. However, coordinated changes across organs do not, by themselves, establish the direction of influence. In a conventional gene knockout, for example, simultaneous expression changes in liver and muscle may result from gene loss acting separately in each organ, indirect effects between organs, or both. PVMouse 3.0 will first describe these joint responses, then combine tissue-specific perturbations, information on perturbation and sampling times, and measurements of relevant signaling molecules to test how perturbation effects may spread between organs. The model will distinguish established physiological mechanisms, experimentally supported relationships whose mechanisms remain unresolved, and predictive associations that still require validation.

Physiological, biochemical and pathological measurements will link multi-organ states to organism-level function and phenotypes. The model will be evaluated to determine which changes in organ state are associated with these measurements and whether combining information from multiple organs improves prediction. Measurements such as blood glucose, which reflect contributions from several tissues, must be interpreted in relation to the states of the relevant organs.

The five tissues and organs initially covered by PVMouse 3.0 form a multi-organ system with a defined scope. The model needs to record this coverage and consider organs and physiological processes not yet included when interpreting systemic changes. Its initial task is to connect multi-organ states with measurable phenotypes, then gradually test those relationships using controlled perturbations. The next section explains how single-gene knockouts and associated physiological, biochemical, and pathological measurements provide evidence for this work.

4.6 Testing Systemic Effects of Genetic Perturbations through Single-Gene Knockouts and Phenotyping

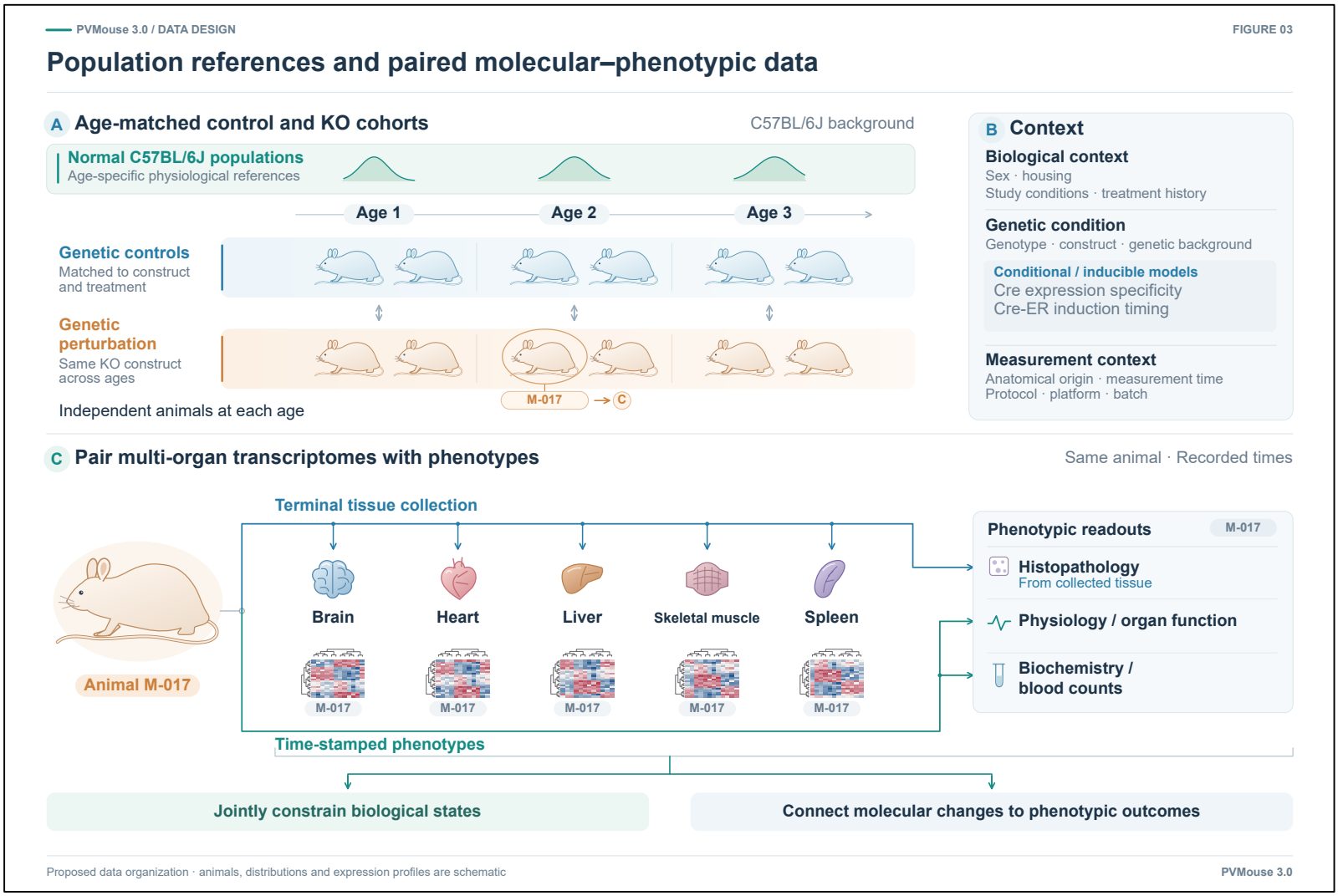
The PVMouse 3.0 project aims to collect multi-tissue bulk RNA-seq data and associated phenotypic measurements from at least 1,000 single-gene knockout mouse lines on the C57BL/6J background, building a genetic perturbation resource that can expand over time. Initial development will focus on a subset with clearly documented genetic constructs, appropriate controls, tissue identities, sample matching and relevant functional measurements. Coverage will then expand to additional genetic perturbations.

Knockout strategies must be described separately. Conventional germline knockouts and conditional knockouts—including tissue-specific and inducible models—differ in which tissues or cells are affected and when gene loss takes effect. Controls must therefore be matched to the genetic construct and experimental procedure. Records should include genotype, actual genetic background, affected tissues or cells, induction conditions where applicable, and sampling time.

Gene loss throughout development may lead to long-term adaptation and compensation (*El-Brolosy et al., 2019*), whereas a knockout induced in adulthood has a different developmental history. The model must therefore account for when gene loss occurs and how long its effects have been present.

Physiological, biochemical and pathological readouts connect molecular changes following genetic perturbation to organ and organism function. Blood glucose, blood lipids, transaminase activity, blood cell counts and tissue injury scores provide relatively low-dimensional readouts with clear physiological meaning. They offer functional reference anchors and constraints for high-dimensional transcriptomic representations. Joint training on knockout and matched control data will pair tissue gene expression with functional measurements from the same animal, helping the model retain molecular features associated with functional changes. These measurements will also help interpret expression changes, identify relevant cell populations and biological processes, and evaluate the effects of genetic perturbations.

Validation must distinguish estimation of an animal's unmeasured tissue or phenotypic readouts from prediction of intervention outcomes. The former uses available observations from the same animal to assess the information shared across measurements; the latter predicts expression and functional changes from genetic background, age, knockout conditions and predefined reference information before the target outcomes are available. Key predictions can then be tested through independent animal experiments, tissue and cell measurements, or rescue experiments directed at candidate mechanisms. Beyond assessing whether predicted relationships hold, these experiments will help identify shortcomings in the model’s representation of biological states, perturbation responses or connections across scales, guiding further refinement.



Population references and paired molecular–phenotypic observations. (A) Normal C57BL/6J populations provide age-specific physiological references. At each age, animals carrying the same KO construct are compared with genetic controls matched to the construct and treatment. Different animals are sampled at different ages. (B) Biological, genetic and measurement context is retained, including genetic background, treatment history, anatomical origin, measurement time, protocol, platform and batch; Cre expression specificity and Cre-ER induction timing are recorded for the relevant conditional or inducible models. (C) Terminal collection provides brain, heart, liver, skeletal muscle and spleen bulk RNA-seq observations from the same animal. Tissue histopathology and time-stamped physiology, organ function, biochemistry and blood counts are linked to that animal. Together, these observations constrain biological states and connect molecular changes to phenotypic outcomes. Animals, distributions and expression profiles are schematic.

4.7 Biological Scope and Extensibility of PVMouse 3.0

As a biological world model MVP, PVMouse 3.0 has an initial scope defined jointly by available data and testable research tasks. Using C57BL/6J mice as a reference, it focuses on five organ modules and combines transcriptomics, available spatial omics, and physiological and

pathological measurements to study age-related states and systemic responses to single-gene knockouts. It aims to establish a connected explanatory chain from genetic perturbations, cellular states, and organ structure to mouse phenotypes. The model's intended use needs to specify genetic background, tissue, age, perturbation method, and predicted outcomes, distinguishing analyses supported by existing observations from extrapolation to new conditions.

The molecular prior layer initially organizes regulatory and functional knowledge around gene nodes. As research needs evolve and additional data become available, it can be extended into a heterogeneous graph that includes RNA, proteins, protein complexes and metabolites. Such a graph would link genes to transcripts, transcripts to protein products, proteins to complexes, and enzymes to metabolic reactions. This would distinguish the RNA and protein products of a gene and represent their functional relationships more explicitly. With corresponding molecular measurements and perturbation data, the expanded graph could provide more detailed priors for multi-omics integration and intervention-response prediction. Whether this added detail is useful will depend on the task and validation results.

Starting with single-gene knockouts, PVMouse 3.0 will examine perturbations that differ in the tissues or cells affected, when their effects begin and how long they persist. Conventional knockouts provide a reference for multi-organ responses to sustained gene loss, while tissue- or cell-type-specific manipulations help connect local changes to systemic outcomes. For a hepatocyte-specific knockout, for example, Cre driver characteristics and recombination validation establish which cells are affected. The cell layer represents these populations, while the virtual liver integrates their states, composition and spatial relationships to support analysis of changes in other tissues and organism-level function. Inducible manipulations additionally define a perturbation start time, allowing the sequence and duration of responses to be investigated.

Drug treatments, dietary changes, and environmental stimuli can further extend the perturbations described by the model. Drug responses depend on dose, exposure within each tissue and duration of action. Diet and environmental conditions can affect multiple organs through changes in nutrient supply, metabolic load, neuroendocrine activity and immune function. A shared biological state space would allow these effects to be represented as changes in cell states, tissue composition, organ function and inter-organ relationships, while preserving information about each perturbation’s targets, dose or intensity, affected tissues and exposure history. Molecular and functional measurements will be needed to establish the relevant response relationships, while observations over time will help distinguish short-term responses, sustained adaptation and recovery after a perturbation ends. Representations and modules developed for genetic perturbations thus provide starting points for drug and environmental tasks, whose suitability must be evaluated in those settings. Chapter 8 describes the additional data dependencies and development arrangements.

The current design relies primarily on transcriptomics to characterize cell states. Proteomics, other omics modalities and corresponding pretrained models can be incorporated as needed. Single-cell protein measurements can supplement cell state representations, while regional or tissue-level proteomics can help calibrate virtual organs. Measurements of protein abundance, post-translational modifications and signaling activity provide information beyond RNA expression. Gene-protein mappings, alignment of paired samples and calibration of organ-level readouts allow new modalities to use the existing interfaces linking cell states, spatial organs and the organism-level system. Transcriptomic and proteomic data can retain separate organ-level representations or be integrated through shared biological variables. These alternatives can be

evaluated to determine whether combining modalities improves predictions of perturbation responses and phenotypes.

Organ coverage can expand by adding nodes to the organism-level network while reusing interfaces for cell states, spatial organization and organ-level observations. Each new organ must be described by its cellular composition, regional structure and functional measures, together with the metabolites, hormones, immune mediators and neural signals it exchanges with existing organs. Adding the kidney, for example, would extend the description of liver-kidney metabolism and clearance, while adding the pancreas would improve representation of glucose homeostasis and its links to liver and muscle. As nodes and observations are added, the model can address a broader range of systemic physiological processes.

These biological boundaries and evidence requirements will determine how modules are defined, what information passes between scales, and how new data contribute to model updates. The next chapter develops the computational architecture of PVMouse 3.0 on this basis.

Chapter 5 | The Computational Architecture of PVMouse 3.0: Physiological References across Ages, Virtual Organs, and Perturbation Propagation

Chapter 4 defined the four biological layers of PVMouse 3.0; this chapter explains how they form a computational system for learning, inference and validation. The central design connects gene relationships, cellular states and spatial organization within organs to a joint multi-organ state. Development begins with a physiological reference across ages for normal C57BL/6J mice under defined experimental conditions, followed by modeling of multi-organ responses to genetic perturbations and their changes over time. These responses may combine effects within affected tissues with interactions among organs. The same state and observation interfaces are intended to support later drug, dietary and environmental tasks as corresponding response data and validation become available. Within this architecture, the first layer supplies gene and pathway priors, the second establishes a cell state reference space, the third organizes cells into regional and organ states, and the fourth connects organs and describes their joint changes with age and perturbation. Each layer performs tasks at its own scale, while shared data conventions, state interfaces and observation models allow evidence from one level to constrain estimates at another.

5.1 The Core Computational Problem: Establish a Physiological Reference across Ages, Then Describe Deviations

PVMouse 3.0 begins by establishing a physiological reference across ages. Rather than a single trajectory that every mouse must follow, this reference describes the biological states within populations at different ages, conditional on strain, sex, housing and other background conditions, and summarizes how these distributions change with age. It provides shared coordinates for normal states and a baseline against which deviations associated with disease, drugs and genetic perturbations can be assessed.

Let b denote the animal's background, including genetic background, sex, housing and other experimental conditions that affect biological processes, and let $u(t)$ denote time-varying perturbations, including persistent genetic changes and treatment conditions at that time. Genetic perturbations are represented by the target gene, editing method, tissues or cells actually affected, onset and conditions of persistence. Compound perturbations are linked to existing experiments through stable identifiers; their representations are learned from expression and phenotype changes relative to matched controls and used for prediction together with dose, route of administration and treatment history. Known targets can provide supplementary priors, while measured responses can themselves establish compound representations.

Measurement conditions, denoted by r_m, include sampling time, anatomical region, assay platform, sample processing and technical batch. At time t, the system state $z(t)$ jointly describes cellular, regional, organ and organism-level states, constrained by available expression, spatial, physiological and pathological data. Observations at scale or modality m, denoted by $y^{(m)}(t)$, may include single-cell or spatial expression, bulk RNA-seq from defined anatomical regions, blood biochemistry, pathology and other functional readouts.

State changes and experimental observations can be represented separately as:

$$z(t+\Delta t) \sim p_\theta\big(\cdot \mid z(t), b, u_{[t,t+\Delta t]}, G; \Delta t\big)$$

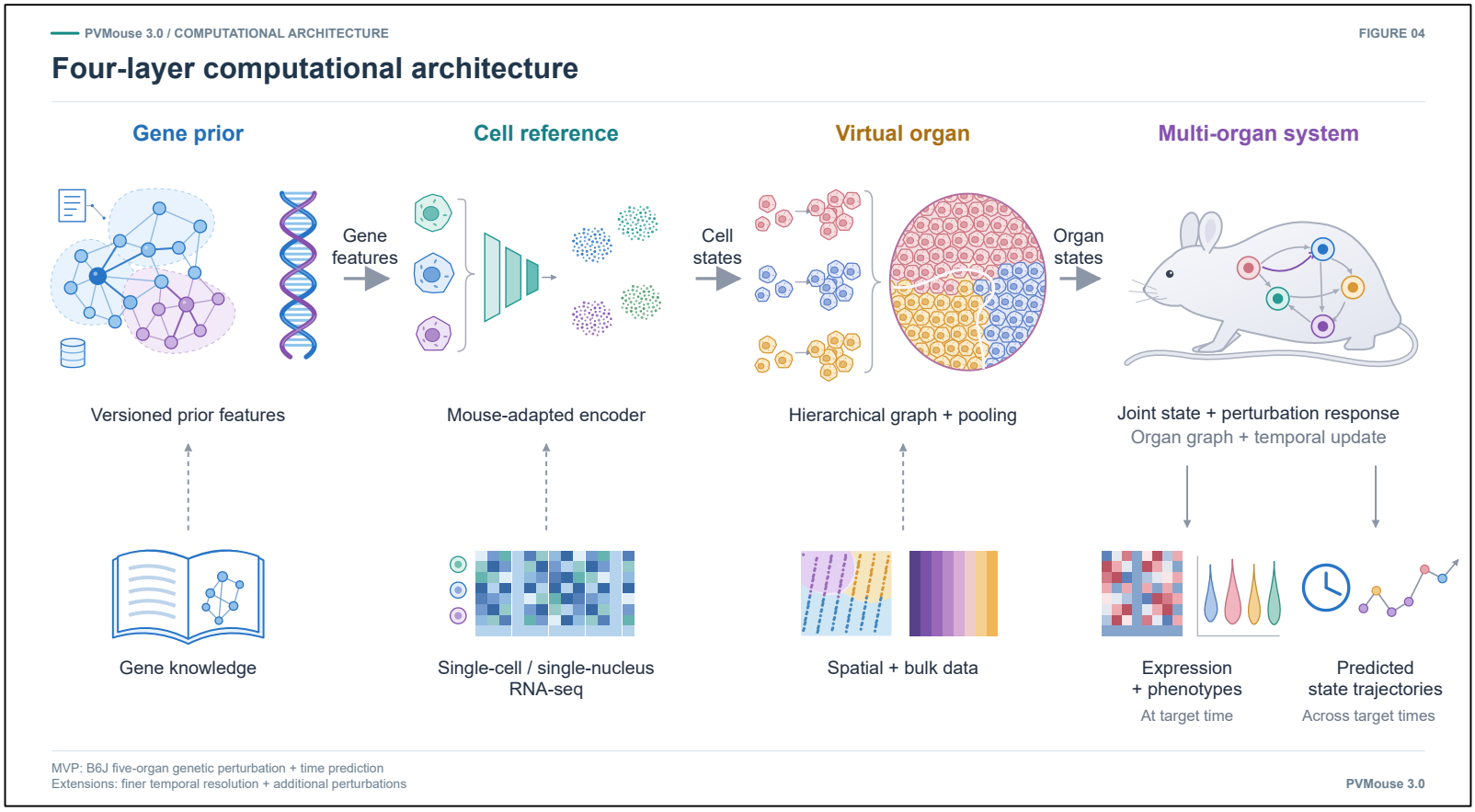
$$y^{(m)}(t) \sim p_{\phi_m}(\cdot \mid z(t), b, r_m)$$

The first equation describes state updates conditional on background, perturbations and structural priors G, while the second specifies the distributions of observations given the latent state. The modality-specific observation model p_{ϕ_m} links latent states to experimental readouts for modality m. Although modalities may differ in their noise, aggregation and missingness mechanisms, they constrain the same multiscale state. The model learns the reference physiological distribution under the unperturbed condition $u(t) = 0$ and estimates how that distribution changes under perturbation. These effects need not be additive shifts: they may depend on current age and state, alter inter-organ relationships or move the system into regions not represented in the reference data.

5.2 Overall Architecture: Two Continually Updated Components and Two Long-Term Stable Frameworks

The four layers combine continually updated knowledge components with modeling frameworks intended to remain stable over time. The first-layer gene priors are updated as molecular relationships and evidence accumulate, and the second-layer cell state references evolve with public atlases, non-public datasets and foundation models. By contrast, the third layer retains the organizational logic of "cells-regions-organs", and the fourth retains the system logic of "joint organ states-temporal change-perturbation propagation".

Here, "long-term stable" refers to computational objects, responsibilities across levels, input-output interfaces and validation objectives, rather than a fixed algorithm, neural architecture or parameter set. The third layer can adopt or combine hierarchical graphs, set models and generative models, while the fourth can use static graphs, state-space models, temporal graph networks or continuous-time models, provided state meanings and validation requirements remain consistent. Information is progressively aggregated: gene priors constrain cellular representations, cell states and composition inform regional and organ models, and organ states enter the joint multi-organ model. Spatial, bulk, physiological and pathological observations calibrate the corresponding levels. Gradients from higher-level prediction losses can reach trainable lower-level parameters, but joint updates should use constrained adaptation to preserve components that have already been validated.



Four-layer computational architecture. Gene-level prior features inform a mouse-adapted cell encoder, whose cellular representations enter virtual organs through hierarchical graphs and pooling. Organ states are then linked in a multi-organ system for joint-state, perturbation-response and temporal modeling. Gene knowledge, single-cell or single-nucleus RNA-seq, and spatial and bulk data provide inputs at the corresponding levels. Outputs include expression and phenotypes at a target time and predicted state trajectories across target times. The MVP covers prediction of genetic-perturbation responses and temporal changes across five organs on the B6J background; finer temporal resolution and additional perturbations are extensions.

5.3 Layer 1: Gene-Level Priors with Documented Provenance

The first layer converts prior knowledge of protein interactions, transcriptional regulation, ligand-receptor relationships, pathways and gene sets into gene-level structural priors. Its outputs include graphs with typed, evidence-weighted relations, gene or pathway representations, and regulatory and functional priors for the cell layer. Gene nodes use consistent mouse gene identifiers and retain the reference genome version; nodes and relations also preserve tissue or cell context, provenance, evidence level and version, distinguishing experimentally supported relationships from candidates inferred only through data analysis.

The purpose is to organize reusable molecular knowledge into a consistent, queryable and updatable foundation, rather than train an all-encompassing gene model from scratch. One implementation could represent transcription factor-target regulation, gene interactions and pathway membership in a graph and derive priors from fixed graph features or a graph encoder. Components supported by stable evidence could be frozen or updated infrequently, while tissue-

specific or newly added relationships could enter the second layer through adapters, regularization terms or additional features.

The value of the first layer must be tested in downstream tasks. If adding priors does not improve cell state identification, organ readouts, or perturbation prediction on independent data, those priors should not be forced into the model merely because their sources are authoritative. This retains the constraining role of existing knowledge while allowing data to suggest new relationships not yet included in knowledge bases.

5.4 Layer 2: A Conditional Cell State Reference Space

The second layer maps high-dimensional single-cell or single-nucleus expression into a shared cell state reference space, using gene and pathway priors from the first layer. Metadata on organ of origin, age (weeks), sex, perturbation conditions and assay platform support adaptation, conditional analysis and validation. Outputs comprise cell state representations, probabilities of cell types and states, expression features of functionally related gene sets, and associated uncertainty.

Pretrained single-cell foundation models can initialize this layer or serve as candidate encoders. scFoundation and models with similar functions can be adapted with lightweight methods such as adapters, low-rank adaptation (LoRA) (*Hu et al., 2022*), or projectors, but no specific model is a prerequisite for the architecture. Model selection needs to address the mouse gene vocabulary, ortholog mapping, genes not covered by the model, and domain shifts across platforms. Candidates should be compared with self-supervised models trained on mouse data, scVI (*Lopez et al., 2018*), and simple baselines under the same data splits. A foundation model is included in a released version only if it provides consistent gains across studies, ages, or downstream organ tasks.

The cell model learns distributions of cellular states conditional on age and other background conditions; cross-sectional data alone do not recover continuous cellular trajectories within the same animal. Training can combine masked reconstruction, cell type or state discrimination, contrastive learning and distribution smoothing between neighboring age groups, informed by age-related expression changes reported in existing studies. Validation focuses on the reliability of these estimates in animals, studies and ages not seen during training and across assay platforms. The resulting references supplied to the third layer retain their conditioning variables, probabilities and provenance rather than constituting a fixed cell atlas.

5.5 Layer 3: From Cellular States to Virtual Organs

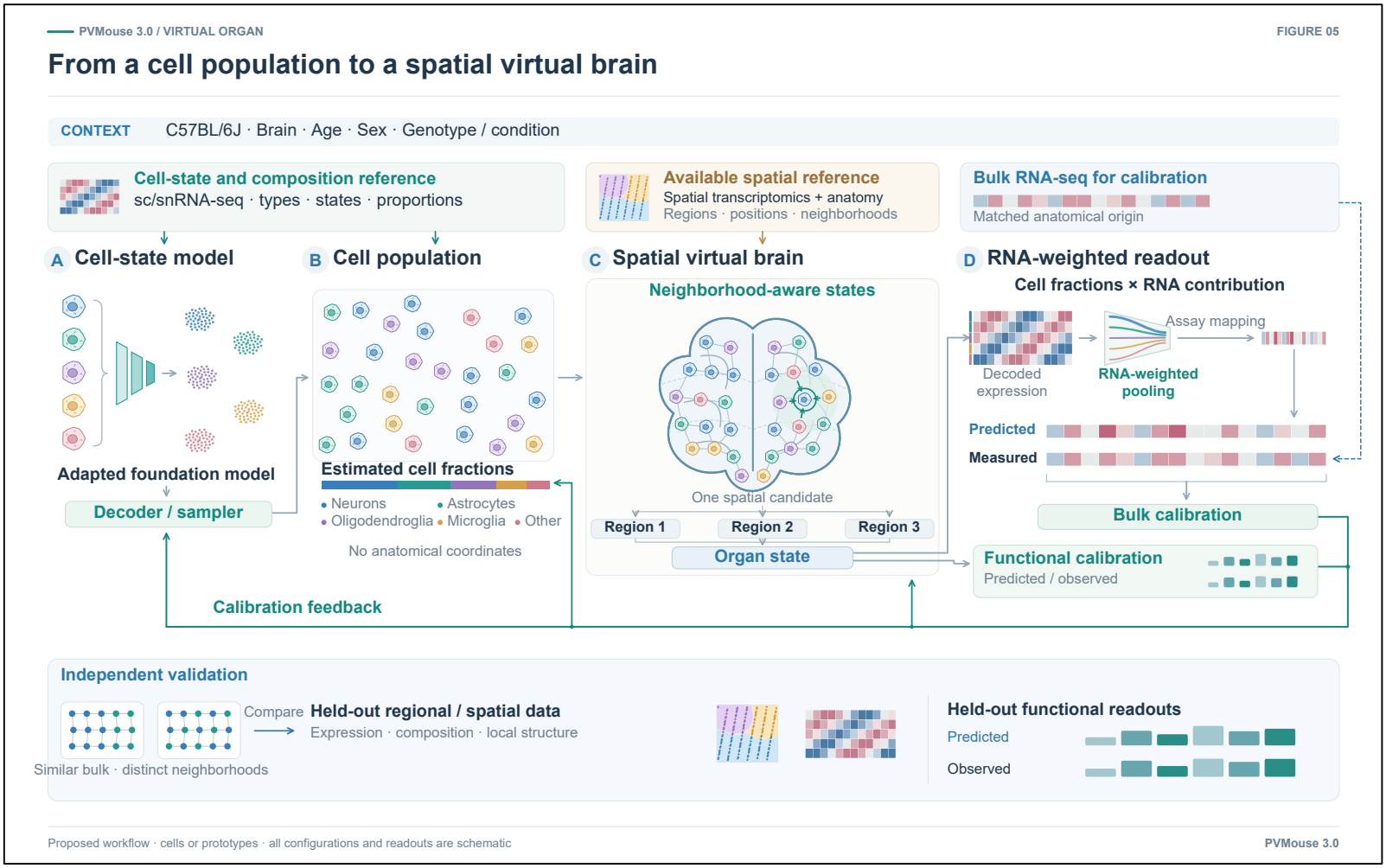
The third layer connects cells to organs by integrating second-layer cell states with cell composition, spatial or anatomical references, and organ-level observations. Regional or position-level states are first formed and then aggregated, following the sequence **set of cellular states → regional and neighborhood states → whole-organ state → measurable organ expression profile**. Because available single-cell, spatial-reference and bulk data generally constrain several possible tissue configurations, a virtual organ does not uniquely reconstruct the cell positions of a particular animal. Instead, this layer learns a conditional distribution over organ states and spatial organization given organ identity, age, sex, perturbation and available observations. It can generate multiple regional structures and cell-neighborhood configurations compatible with the evidence while distinguishing observed features from those that remain unidentifiable.

Possible implementations include hierarchical graphs, set models with positional encoding, regional attention pooling, optimal transport and conditional generative models for cell-region-organ relationships. Cell nodes or prototypes carry cellular states, regional nodes encode local composition and spatial structure, and organ nodes summarize the overall state. Spatial atlases constrain regional organization and neighborhoods, measured bulk RNA-seq calibrates overall expression, and pathological and functional measurements test the biological relevance of the resulting organ states.

An organ state must extend beyond a bulk expression vector to describe cell composition, regional states, cell-neighborhood relationships and coordinated expression changes associated with organ function. Although bulk expression provides an important measurement and alignment target, deriving it from cellular states requires accounting for RNA content, cell size, sampled anatomy, sequencing platform and the way cellular contributions combine in the measurement. Reliability must therefore be assessed jointly against cell composition, spatial structure, bulk expression and independent functional measurements.

Each organ module will first use public single-cell, spatial or anatomical reference, and bulk data from normal C57BL/6J mice to connect cellular states, regional organization and overall expression. A data-rich organ can serve as the spatial prototype, whose interfaces can then be reused to construct the remaining normal organ modules and connect them into a multi-organ reference model. This initial stage evaluates tissue and expression reconstruction under normal conditions; perturbation responses are learned jointly after the normal multi-organ model has been established.

The third layer ultimately supplies the fourth with the state of each organ: cell-type composition, regional states, functionally relevant gene expression changes, predicted overall expression profiles, and uncertainty in these estimates. The fourth layer uses these organ states to analyze how multiple organs change together.



From a cell population to a spatial virtual brain. The target context specifies C57BL/6J, brain, age, sex and genotype or condition. (A) An adapted foundation model and conditional decoder or sampler use cell-state and composition references. (B) The resulting cell population has estimated cell fractions but no anatomical coordinates. (C) Available spatial transcriptomic and anatomical references constrain regions, positions and neighborhoods, producing neighborhood-aware states, regional representations and an organ state. The illustrated brain is one spatial candidate. (D) Cell fractions and RNA contributions enter RNA-weighted pooling of decoded expression; assay mapping connects the readout to measured bulk RNA-seq from a matched anatomical origin. Bulk and functional calibration provide feedback. Independent validation considers configurations with similar bulk expression but distinct neighborhoods, using held-out regional or spatial expression, composition and local structure, and held-out functional readouts. Cells or prototypes, configurations and readouts are schematic.

5.6 Layer 4: Joint Multi-Organ States and Temporal Evolution

The fourth layer connects virtual organs into an organism-level state model, representing each organ as a node with its third-layer state as node features. Edges encode known circulatory, metabolic, immune, neuroendocrine and other physiological relationships. Blood metabolites, hormones, cytokines and other systemic mediators can serve both as explicit variables for inter-organ information transfer and as measurable organism-level observations.

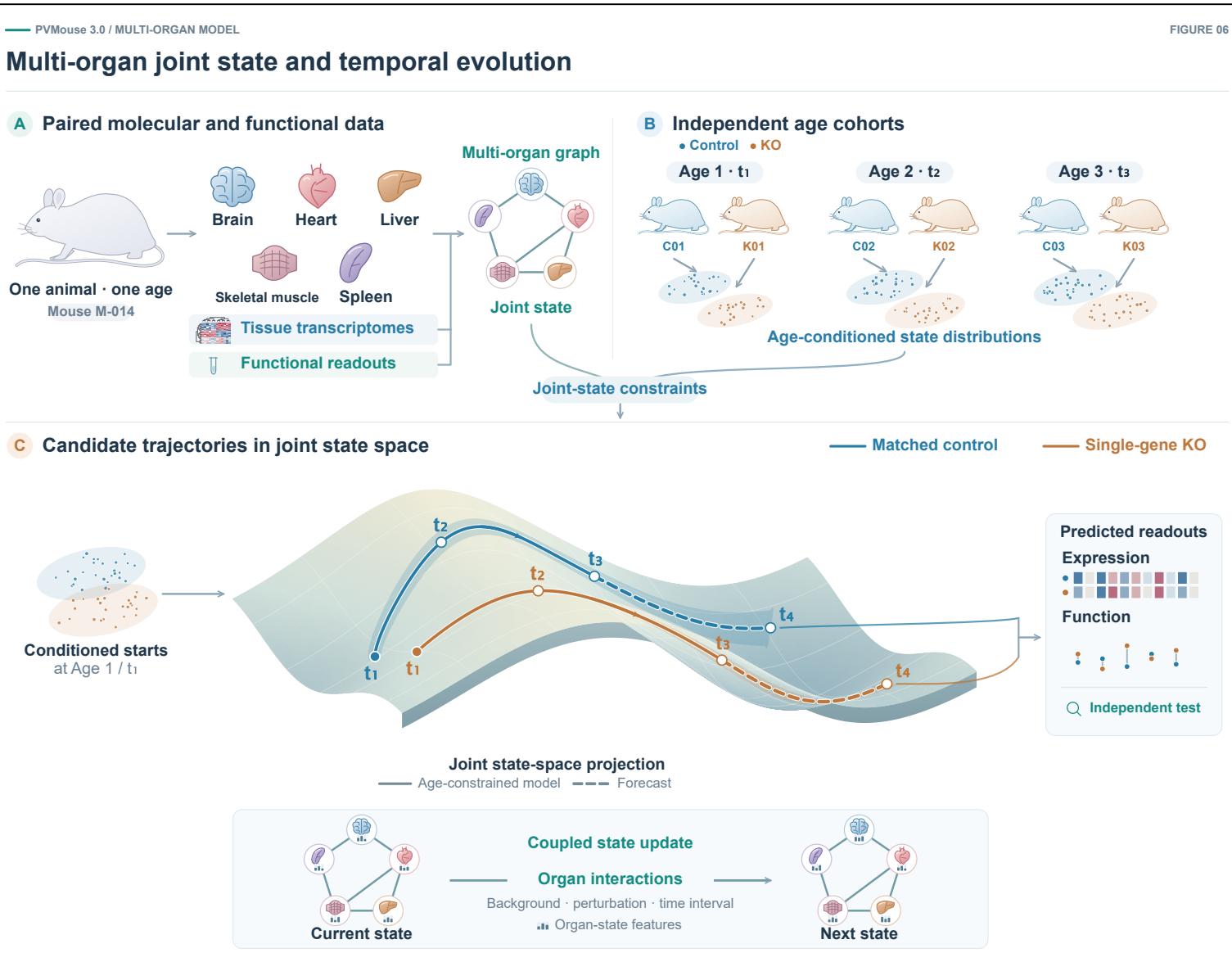
This layer initially connects organ states in normal C57BL/6J mice, using available public multi-tissue data and prior knowledge of inter-organ physiology to establish a reference model. Multi-tissue transcriptomes, blood biochemistry, blood counts and pathological readouts from the same animal jointly constrain organ states and the relationships between molecular and functional readouts, while observations across ages define population-level physiological references. Once the reference model has undergone initial validation, data from genetic perturbations, initially single-gene knockouts with appropriate matched controls, are incorporated to model multi-organ responses. Early evaluation can focus on brain-related expression and phenotypes while retaining other organ states and interactions, because brain changes may reflect local responses, influences from other organs or systemic feedback that must be interpreted together.

Perturbation modeling starts from a state consistent with the genetic construct, age and treatment history. The documented tissues or cell populations in which a perturbation acts define its direct inputs to the model. Organ states are then updated jointly, accounting for local responses and feedback among organs. Predictions of response timing draw on temporal observations and the assumed dynamics. In later compound-response tasks, measured responses and available priors can support predictions of which tissues respond and hypotheses about possible sites of action. These hypotheses distinguish direct action from downstream effects and are assessed using available exposure, target-engagement or other mechanistic evidence, with uncertainty retained for subsequent experimental testing. Estimates of the scope of action include uncertainty and serve as candidate explanations for later experimental validation. For conventional knockout, the model can learn associations between the construct and whole-body responses, but endpoint data alone cannot determine which organ influenced which other organ. Cross-organ relationships should be distinguished by evidence level: covariation, support from whole-body perturbations, support from tissue-specific interventions, and direct mediator evidence. Directionality requires further testing with conditional perturbations, temporal order, and mediator measurements.

For data with irregular sampling times or event inputs, state-space models, graph recurrent models, or temporal graph networks can be used. Continuous-time implementations such as graph neural ordinary differential equations (ODEs) (*Poli et al., 2019*) are used only when the data support continuous-time modeling and the model improves on baselines. A general organ state update can be written as:

$$z_i(t+\Delta t) \sim p_\psi\left(\cdot \,\middle|\, z_i(t), \{z_j(t)\}_{j\neq i}, b, u_{[t,t+\Delta t]}, G_{organ}; \Delta t\right)$$

Here, $z_i(t)$ is the state of organ i, and G_{organ} is an organ graph with relation types and evidence levels. The equation states that an organ's next state depends jointly on its current state, other organ states, background, perturbations, and the time interval. It describes a conditional distribution to be tested, not a predetermined propagation path.



Multi-organ joint state and temporal evolution. (A) Tissue transcriptomes and functional readouts from one animal are linked through a graph of brain, heart, liver, skeletal muscle and spleen states. (B) Independent control and KO cohorts at three ages provide age-conditioned state distributions and joint-state constraints. (C) Matched-control and single-gene-KO trajectories begin from their respective condition-specific states at age 1 and are shown in a shared joint-state projection. The three ages correspond to t_1, t_2 and t_3, while the dashed segment from t_3 to t_4 denotes the forecast. Predicted expression and functional readouts are linked to independent testing. Current organ-state features and their interactions enter a coupled update under specified background, perturbation and time interval to produce the next state. Model relations and candidate trajectories are conceptual.

5.7 Training and Inference: Two Data Flows within One Framework

Training develops the layers according to their dependencies. The first organizes and encodes gene-level priors, the second establishes conditional cell state references from single-cell data, and

the third learns regional and organ states from single-cell, spatial, bulk and functional observations. The fourth first learns joint multi-organ distributions from paired tissues and then uses observations across ages, controlled perturbations and available longitudinal measurements to test temporal changes and propagation relationships. Following independent validation of each layer, adapters or constrained parameter updates provide cross-layer alignment.

Normal samples, samples from knockout mice and their appropriate genetic controls enter the same framework, with knockout construct, age, sex and experimental context supplied as conditioning inputs. Joint training preserves the recorded pairing between multi-tissue transcriptomes and physiological, biochemical and pathological measurements from the same animal. For a knockout-response prediction task, the target joint conditional distribution can be written as:

$$p_\theta\left(x_{multi\text{-}organ}, y_{physiology}, y_{pathology} \mid g, b, t\right)$$

Here, g denotes the genetic construct, b the animal background defined above, and t the task-relevant time; the three observed variables describe multi-tissue expression, physiological measures and pathological outcomes. Their conditional joint distribution is modeled through a shared mouse state, from which the corresponding observation models predict each readout for comparison with measurements. Expression and phenotype differences between knockout mice and their matched controls also serve as training targets, linking genetic changes to organ responses and functional consequences.

Training errors are computed separately for each observation type according to its data type, measurement scale and noise level, then used jointly to train the state representation. Masks exclude error terms for missing observations while retaining records of why data are missing. Phenotypes contribute through organ- and organism-level prediction tasks and, during controlled cross-layer alignment, can also constrain cellular representations requiring adaptation.

Inference begins with a clearly defined biological query specifying strain, sex, age, environment, target organ and time, and any genetic or compound perturbations. The model estimates the initial state from the animal's background, treatment history and available observations; this state may be normal or already altered by genetic changes or treatment. Genetic perturbations are described by the target gene, editing method, scope, and starting time: effects already present before the simulation begins are included in the initial state, while the genetic conditions responsible for them remain active during subsequent computation. For compound perturbations, stable identifiers retrieve effect representations learned from existing experiments, which the model combines with dose, route, and dosing schedule to predict responses from the current state. This computation must account for both the effects of previous treatment on the initial state and the order, intervals, and durations of subsequent treatments.

The third layer generates or updates regional and overall organ states, which the fourth combines with perturbation information to predict multi-organ responses at the target time. Documented sites of direct action enter as conditions; where these are unknown, the model predicts tissue responses and retains possible sites of action as hypotheses. The normal physiological reference provides a basis for interpreting these states and changes, but **"first reconstruct normal states, then simulate deviations" specifies the development sequence, not a requirement to**

begin every prediction from a normal state. Each inference starts from the current genetic conditions, treatment history and available observations; supplied measurements update the posterior distribution over possible states and thereby revise the initial estimate.

Final outputs are predictions under the specified conditions, with their scope of applicability and uncertainty. They include regional or organ states, measurable molecular and functional readouts, other organs that may be affected, candidate propagation relationships, and time windows needing experimental tests. For conditions poorly covered by training data, with substantial missingness across modalities, or clearly outside the data distribution, the system should lower confidence or decline to simulate beyond the evidence.

5.8 Layer-Specific Validation

The complexity of PVMouse 3.0 is justified only by consistent gains over simple baselines. Evaluation at the first layer asks whether gene priors improve generalization in cell state or perturbation tasks; at the second, whether cell types, states and expression features of functionally related gene sets are reliably represented in independent studies and at ages not seen during training. The third layer tests joint agreement of cell composition, regional expression, neighborhood statistics, bulk readouts and organ function with independent observations, while the fourth tests whether predicted multi-organ responses, genetic perturbation effects and temporal changes are reproduced in independent data.

Comparisons use identical data splits and specify the observations and conditions available to each model. To distinguish the contribution of information from that of model architecture, predictions are compared before and after adding information, and simple and complex models are compared using the same available inputs. Validation addresses three questions:

- Do cellular and spatial information improve predictions of organ gene expression and function? Compare models using bulk data alone, adding cellular representations, and further adding spatial organization to test whether each addition improves prediction of organ expression and related function. Comparisons with and without molecular priors further assess the contribution of prior knowledge. Where the data allow, comparisons will also distinguish the contribution of anatomical region information from that of cellular neighborhood structure.
- Does jointly modeling tissues from the same animal provide additional information? Compare the joint model with independent models for each organ and methods using background conditions alone to test whether information across organs improves estimates of target tissues and phenotypes. For tasks that use measurements from some tissues to predict others, also compare with simple regression models using the same observations and conditions. This tests whether complex modeling of organ relationships provides additional benefits.
- Do genetic perturbation and temporal modeling improve prediction of systemic responses? Predict expression and functional changes in KO relative to matched controls and compare with statistical models using only background and perturbation conditions, bulk models, and other baselines. This tests whether the multiscale design adds predictive value. In temporal tasks with defined starting points and subsequent observations, compare

temporal models with age-conditioned static predictions and simple temporal models to test whether state updates improve predictions at later time points.

The contributions of cross-layer training designs, such as phenotype supervision, are assessed by removing the corresponding training constraints. These comparisons and the independent observation tests above jointly inform whether components should be retained, adjusted, or updated.

Data splitting uses the animal as the basic independent unit. Tissues, blood, pathology, and repeated observations from the same animal enter the same set, while spatial sections and regional data from the same sample remain grouped (*Lazic, 2010*). Test tasks are defined according to the generalization goal, including individuals, studies or batches, ages, knockout constructs, and target genes not seen during training. Age generalization distinguishes interpolation within the observed age range from extrapolation beyond it. Genetic perturbation generalization can also group target genes by family to test transfer of knowledge about related genes. Litter, cage, or kinship relationships are handled through grouped splits in the relevant evaluations. Public data potentially used in foundation model pretraining should be marked for possible leakage. Model selection uses a development set, while final conclusions use an external challenge set locked in advance to preserve evaluation independence.

Uncertainty outputs are tested alongside prediction errors. For continuous expression and functional readouts, evaluation examines prediction interval coverage and width on independent data and reports performance separately across time horizons and generalization conditions. For classification readouts, it tests whether predicted probabilities agree with observed frequencies. When low-confidence warnings or abstention mechanisms are used, the fraction of predictions retained and their errors are reported together to assess whether the mechanism identifies unreliable simulations.

These architectural and validation principles provide a computational framework in which multiscale observations contribute to shared state estimation and response prediction. Continued improvement depends on how new data enter the model and how experimental tests guide revisions to states and relationships. Chapter 6 develops this continual learning and experimental feedback mechanism, Chapter 7 presents research applications, and Chapter 8 sets out the staged development path for the C57BL/6J five-organ model using specific datasets and reusable components.

Chapter 6 | From Data Accumulation to Knowledge Growth: Continual Learning and the Research Feedback Loop of a Biological World Model

Once the initial model has been built, new experiments will continue to provide observations of biological states, responses to perturbations and functional outcomes beyond those represented in the original data. These findings may support existing predictions or expose gaps in the model's representation of cell populations, inter-organ relationships or temporal processes. Updating PVMouse 3.0 will therefore require assessing which existing relationships remain supported, which need revision and what additional observations would be most valuable.

The model can also inform subsequent data collection. Its predictions can help researchers compare competing hypotheses, identify gaps in observations of particular tissues, time windows or biological processes, and prioritize experiments. Results then return to the model as evidence for further learning, connecting data accumulation, model updates and experiment selection in a computational-experimental feedback loop.

This feedback loop offers two potential benefits: it allows data from different studies to be reused in learning relationships among states and intervention responses, and it can help direct limited resources toward more informative experiments. Whether model updates improve predictive accuracy and reliability, and whether model-assisted selection improves research efficiency, must be evaluated separately using independent data, subsequent experiments and appropriate comparisons. The following sections examine data reuse, learning from new observations, experiment selection and their contributions to research efficiency.

6.1 Data Accumulation and Reuse across Studies

Although biological experiments are usually designed to test a hypothesis or answer a specific question, the data they generate can support research beyond their original purpose (*Lachmann et al., 2018*). A gene knockout study may measure gene expression in several tissues alongside blood biochemistry and pathological phenotypes. These observations can provide evidence about the gene's function, coordinated changes across tissues, links between gene expression and function, and responses to related genetic perturbations across different gene-edited strains. Standardized data descriptions and consistent state representations will allow PVMouse 3.0 to reuse these observations across multiple research tasks.

For example, a study focused on liver changes after a gene knockout may also collect skeletal muscle, spleen and blood biochemistry data. Once incorporated into PVMouse 3.0, these observations can reveal which tissues respond, whether hepatic expression changes accompany altered blood lipids, and how consistently organs respond across animals. When another gene is subsequently studied, comparing the resulting tissue expression profiles and phenotypes can suggest shared biological processes or distinct routes of action. This extends the value of the original experiment from its specific research question to broader analyses of the biological system.

Further information can also come from stored tissue samples that have not yet been analyzed. Collecting and appropriately storing additional tissues at a planned experimental endpoint preserves opportunities for future analyses. In the example above, researchers might also store adipose tissue from the same animals for later studies beyond the initial five-organ scope. If

subsequent analysis suggests that adipose tissue could help explain differences in blood lipids, additional assays could be performed, provided the stored samples remain suitable. The new results could then be linked to liver gene expression and blood lipid measurements from the same animals, adding multi-organ information and potentially reducing the need for further animal experiments.

Reusing data across studies requires documentation of the original experimental context, measurement conditions and the links between samples and their source animals. Data from different studies and animals can jointly constrain the distribution of organ states under comparable conditions. Measurements from the same animal provide additional information about how states in different tissues occur together. Negative results are also informative: when the perturbation is confirmed to be effective and the assays are sufficiently sensitive, they help identify conditions in which an expected effect was not observed.

Reanalysis of existing data and additional measurements on stored samples can therefore yield new information from earlier experiments. The next section discusses how this information can be used to revise the model and improve its predictions.

6.2 Learning and Updating Models with New Data

When new data become available for PVMouse 3.0, the first step is to determine which problems they can help the model address. They may provide observations of cellular or disease states not previously covered, or help explain errors in predictions of tissue responses or organism-level phenotypes. Where predictions already exist, comparing them with the new observations can reveal the conditions under which errors arise and help identify which model components need adjustment.

The value of accumulating data depends on both the number of samples and the information they add. Repeated experiments under the same conditions help estimate variation between individuals and variability in measurements. Sampling across a wider range of ages, sexes and perturbation conditions allows the model to learn about states and intervention responses absent from the existing data. Adding protein, metabolic or functional measurements to tissue transcriptomic data may help distinguish states with similar gene expression profiles but different functional outcomes. Measurements at additional time points provide information about the response sequence, duration and recovery. Together, these data help the model describe biological variation, learn how systems respond to perturbations and predict subsequent changes.

Returning to the example in the previous section, an initial model may reconstruct liver gene expression profiles after a gene knockout but fail to predict changes in blood lipids accurately. Additional measurements from adipose tissue can be combined with the existing liver expression and blood lipid measurements from the same animals to train the model to learn relationships among these observations. If animals with similar liver transcriptomes differ in their adipose tissue states and blood lipid levels, the additional information may explain variation the model could not previously distinguish. Independent data are then needed to test whether including adipose tissue information actually improves blood lipid predictions. If errors persist, the next step is to consider whether measurements from other tissues, additional metabolic measurements or revisions to the modeled relationships are needed.

How the model is updated depends on what the new data contribute. New evidence about biomolecular interactions can inform revisions to prior assumptions about those relationships, while additional samples of the same type can support further training of existing modules.

Incorporating previously unmodeled tissues, measurement modalities or biological processes may require additional inputs, state variables or prediction outputs. For a task with a clearly defined boundary, adaptation can initially focus on the relevant modules and their connections. Broader joint retraining can follow when data coverage expands substantially or when these targeted adjustments fail to improve predictions across multiple biological levels.

Retraining combines new data with representative existing data, allowing the model to continue training on established tasks as it learns from new conditions. For each update, the data used, the components modified and the associated tasks are recorded, and the model version is saved. This allows results to be reproduced and performance to be compared before and after the update.

Updated models must be evaluated on data used for neither training nor model selection in any version being compared. Under the same test conditions, the new version is compared with the previous version and appropriate simple baseline models. The evaluation assesses whether predictions improve on the target tasks, whether uncertainty estimates are reliable and whether performance on existing tasks has declined. Improvements confirmed by independent evaluation can inform the next round of applications and experiment selection. Remaining prediction errors help determine what data to collect next and how the model should be revised.

6.3 Model-Assisted Experiment Selection

As data accumulate, PVMouse 3.0 can provide more detailed comparisons of biological states and predictions of responses to perturbations. These outputs can help researchers interpret existing experiments and plan subsequent studies. By comparing a wide range of candidate conditions, identifying where predictions differ and where additional observations could fill important gaps, and suggesting interventions worth testing, the model can help researchers narrow down which experiments to perform.

Experimental recommendations should address a clearly defined research question. For intervention screening, model predictions can help researchers compare potential functional improvements, effects on other organs and the uncertainty associated with these outcomes. For mechanistic research, the model can identify conditions under which competing explanations lead to different predictions. For model development, it can identify gaps in the available observations that may affect several tasks. These objectives can be pursued together or separately.

For example, a conventional gene knockout may alter both liver gene expression and blood lipid levels. The model may suggest different explanations: loss of the gene in the liver contributes to changes in blood lipids, or responses in other tissues affect both liver expression and blood lipids. To investigate the contributions of individual tissues, researchers can ask the model to predict liver gene expression and blood lipid levels under two interventions: knocking out the gene only in hepatocytes in adulthood, or only in another candidate tissue. Predictions for each intervention are then compared with those for its corresponding control condition. If the competing explanations lead to clearly different predictions for a particular experiment, researchers can prioritize that experiment and plan tissue collection, blood assays and sampling times accordingly. The results can then be used to reassess the contributions of individual tissues and revise the relevant predictive relationships.

The model can also help researchers decide what to measure and when. Suppose changes in tissue gene expression are observed soon after treatment, while functional abnormalities appear

later, but intermediate observations are insufficient to establish how the response develops. Using the available data, the model can propose possible trajectories linking these stages and predict when the relevant readouts would change along each trajectory. One explanation might place metabolic alterations well before dysfunction, while another predicts that they occur close to its onset. Researchers can prioritize intermediate time points at which these predictions differ most and select metabolic or functional assays based on biological relevance and technical feasibility. The resulting observations can test the proposed trajectories and help connect early responses with later phenotypes.

Experiments are prioritized according to their scientific value, expected benefit and resource requirements. An intervention with a large predicted effect may warrant testing, but an experiment with a small expected effect can also be valuable if its results could rule out an important explanation. Model uncertainty can help inform this decision, but researchers must consider whether it reflects limited evidence, variation between individuals or measurement noise, and whether the proposed experiment could reduce it. These considerations are weighed alongside cost, the time required to complete the experiment, measurement resolution and the potential for the results to influence subsequent research decisions.

6.4 The Computational-Experimental Feedback Loop and Research Efficiency

The PVMouse 3.0 feedback loop progresses through successive rounds of research. In each round, researchers identify questions based on the current model and research objectives, plan experiments, organize the results and update the relevant model components before generating new predictions. Recording the model version, inputs and principal predictions before each experiment, followed by observed outcomes and quality information, preserves a continuous account of predictions, revisions and subsequent decisions.

In the gene knockout example, an initial round of measurements from multiple tissues in the same animals may reveal changes across the liver, adipose tissue and blood lipids. Integrating these observations can help the model generate alternative explanations for the contributions of different tissues. A second round of experiments, using tissue-specific conditional knockouts or measurements at additional time points, can help distinguish these explanations. Once the new data have been used for retraining, the revised predictions are tested using data from independent animals or conditions not included in training. If the model predicts responses more accurately under defined conditions, it can support subsequent comparisons of related perturbations. Remaining prediction errors can help identify where additional observations or changes to the model are needed.

Model-assisted experiment selection aims to reduce redundant experiments and direct limited resources toward testing key predictions and exploring new questions. These benefits, along with those of data reuse, must be evaluated in the context of specific research tasks. For example, under comparable resource budgets, model-assisted screening can be compared with existing workflows on two measures: how many selected genes, compounds or treatment regimens meet predefined criteria in subsequent experiments, and how long it takes to achieve the research objective. In mechanistic studies, evaluation can focus on whether the selected experiments distinguish competing explanations and narrow the set of hypotheses that remain to be tested. The benefits of data reuse can be assessed by tracking how existing data contribute to later studies and how much

repeated data collection or processing is avoided. Task-appropriate measures can thus assess research efficiency in terms of the resources invested, the time required and the results achieved.

Improvements in predictive performance and the benefits of model-assisted experiment selection must be evaluated separately. Whether more accurate predictions lead to more valuable experiments must be judged from the results of those experiments. Success in a single experiment also needs to be considered across multiple research rounds to assess whether the recommendation method remains useful.

Each usable version ultimately has a record of application-relevant capabilities: covered biological backgrounds, tissues, perturbations, and time ranges; the molecular and phenotypic outcomes it can predict; and the independent evaluations supporting those capabilities. New data help expand what the model can do, evaluation establishes the conditions under which it can be used, and practical research generates information for the next round of learning. These records provide the basis for the applications described in Chapter 7, helping researchers choose suitable models and task conditions, translate predictions into candidate interventions and experimental plans, and feed new evidence back into the same cycle.

Chapter 7 | From State Prediction to Intervention Selection: Applications of PVMouse 3.0

The state and response models of PVMouse 3.0 are intended to support five areas of research: identifying disease models that may be suitable for further testing of a candidate compound; designing intervention combinations and dosing schedules; identifying tissues that may show changes of concern after treatment; identifying potential therapeutic targets and how to modulate them to improve disease-related outcomes; and selecting measurements to monitor drug effects or tissue injury. All five applications rely on estimates of current biological states and predictions of intervention outcomes, but each involves different comparisons, explores different variables and requires different experimental evidence for validation.

This chapter outlines how these applications can be studied within a shared modeling framework. For a defined biological context and initial state, the model predicts changes in gene expression and function across tissues under different intervention conditions. These predictions help researchers compare candidates and select those that best meet the study objectives. PVMouse 3.0 will initially focus on genetic perturbations, while response relationships for compounds and combination treatments will need to be learned and tested as the corresponding data become available. Each section presents the biological rationale, computational approach and expected outputs for an application, showing how model predictions can inform candidate selection and the design of follow-up experiments.

7.1 From the Normal C57BL/6J Reference to Digital Twins of Disease Models

PVMouse 3.0 follows the approach of "first reconstruct normal states, then simulate deviations." Multi-tissue data from normal C57BL/6J mice across ages establish a physiological reference for comparing gene expression profiles and organism-level phenotypes across sexes, ages and experimental conditions. Data from an expanding collection of single-gene knockout lines, including tissue-level bulk RNA-seq and phenotypic measurements, provide examples of how defined genetic interventions alter biological states. Whereas normal-reference data help characterize physiological states across ages and conditions, perturbation data help the model learn how states change in response to intervention. Together, they provide a basis for constructing computational representations of disease models.

Within PVMouse 3.0, gene-edited disease models on the C57BL/6J background can be represented in terms of their responses to defined genetic perturbations relative to the normal physiological reference. Model inputs specify the edited gene, editing method, targeted tissues or cell types, and timing of the edit's effects. These inputs must also capture differences in biological history: germline edits affect the system throughout development, whereas inducible edits take effect following induction at a specified time.

Incorporating a disease model begins by combining information about its genetic construct and experimental conditions with multi-tissue bulk RNA-seq data across ages and biochemical, physiological and pathological measurements. The data must preserve the links between tissue measurements and phenotypes from the same animal, together with information about matched wild-type or other appropriate genetic controls. Through the interfaces described in Chapter 5, these observations are mapped into the existing biological state space to update representations of perturbations and tissue states, along with the mappings from biological states to phenotypic

measurements. The disease label identifies the model's intended research use, while the genetic construct and experimental conditions define the biological system being represented. Together, these data provide the observational basis for training and calibrating a digital twin of the disease-model strain.

By integrating disease mouse models on the C57BL/6J background into the same biological state space, PVMouse 3.0 can reuse biomolecular priors, cell state references and organ modules across models. Tissue and phenotype data from each disease mouse model are then used to learn its specific perturbation responses. The shared framework can continue to expand as data from new disease models are added.

In this white paper, a digital twin of a disease-model strain is a multi-tissue state model constrained by a specific genetic construct, experimental conditions and measured data. It describes the distribution of biological states and how they change under those conditions, supporting comparisons of tissue-level gene expression and functional outcomes before and after intervention. This digital twin represents a strain under specified conditions; cross-sectional data from different ages initially constrain population-level patterns of change. Predicting an individual animal's trajectory additionally requires its initial observations, intervention history and a temporal model, with predictions tested against longitudinal measurements.

For gene-edited disease mouse models whose data have not yet been incorporated into PVMouse 3.0, the genetic conditions can still be encoded to generate extrapolated states using learned perturbation relationships. These predictions have different evidential support from digital twins trained and calibrated with observations from the corresponding disease-model strains. The distinction must remain explicit in the outputs, with additional data used to test and refine the extrapolated disease states and intervention-response predictions.

7.2 A Shared Computational Basis for Five Applications

All five applications begin with an estimate of the current biological state, predict multi-organ expression and functional changes under specified interventions, and compare candidates according to the research question. Genetic interventions are specified by the target gene, construct strategy and edited tissues or cell types; drug treatments by the candidate compound, dose and dosing schedule. The duration of an intervention and previous treatment history also inform predictions. When simulating treatment, the model retains the disease model's original genetic construct and uses its current biological state as the starting point for predicting the effects of additional interventions on tissue gene expression and organism-level phenotypes.

A candidate compound can be represented through its observed biological effects without requiring disclosure of its chemical identity. Researchers can use consistent coded identifiers for small- or large-molecule candidates and provide the dose, route of administration, formulation, treatment history, sampling time, controls and corresponding measurements. These data allow the model to learn the response patterns associated with each candidate under different conditions. Chemical names and structures are therefore not required for this observation-based representation, which captures information about the candidate's measured biological effects.

When a candidate compound is evaluated in a digital twin of a disease-model strain, the model predicts its effects using the disease state, the candidate compound's learned representation, the dosing schedule and the prediction time point. Observations of drug effects from healthy mice provide a starting point, but disease can alter target tissue states, inter-organ relationships and

treatment responses. The model therefore needs to predict drug effects in the context of the disease state, rather than simply add disease-related changes to the drug-related changes observed in wild-type mice; such addition can serve only as a simplified estimate. GEARS (*Roohani et al., 2024*) and CPA (*Lotfollahi et al., 2023*) provide precedents at the cellular level for predicting combinations of genetic perturbations and responses to combinations of conditions such as drug identity and dose, respectively. PVMouse 3.0 aims to extend these ideas to an organism-level mouse model constrained by multi-organ observations and phenotypic measurements.

Predicting combined interventions also requires accounting for how one intervention changes the biological conditions under which another acts. An earlier treatment may alter cell states or feedback processes and thereby change the response to a later treatment. The model will predict responses from the current state and intervention inputs, using data from combination treatments and treatments in disease models to learn the relevant interactions. Single-drug data and observations from wild-type mice provide an initial basis, but do not fully determine responses to new combinations or treatments in new disease contexts. Additional experiments are needed to test these predictions and refine the modeled relationships.

All five applications use two reference comparisons. The first is the untreated disease state under the same conditions and at the same time, which allows predicted treatment effects to be assessed relative to the natural disease course. The second is the normal C57BL/6J state at the same age and sex, which provides a physiological reference. Evaluation prioritizes disease endpoints and tissue-level readouts with clear functional meaning, with gene expression profiles providing supporting context. Distances between state representations must be interpreted through the corresponding differences in gene expression and function.

7.3 Application 1: Exploring and Recommending Potential Indications for Candidate Compounds

When testing a candidate compound in a disease mouse model, researchers can also conduct dose- and time-response experiments in wild-type C57BL/6J mice matched for sex and age, collecting multi-tissue transcriptomic data and organism-level phenotypic measurements. PVMouse 3.0 will use these observations to learn a representation of the compound's effects, then predict its responses in digital twins of multiple disease-model strains to identify potential indications for further investigation and experimental testing. Treatment-response data from the original disease model can also be incorporated, providing direct evidence of the compound's effects in a disease context.

Different diseases may share metabolic, inflammatory or tissue repair processes, and a compound may influence several of these processes. An effect observed in the original disease model may therefore be worth studying in other models with related abnormalities. PVMouse 3.0 can use the cell and tissue states represented in each disease model to predict whether treatment might improve relevant functional endpoints and assess accompanying changes in other tissues. Shared biological processes provide clues to potential indications, while each disease's initial state and treatment conditions determine its specific response.

The analysis begins by specifying the candidate disease models, ages, experimental conditions and biologically meaningful primary endpoints. For each digital twin of a disease-model strain, PVMouse 3.0 will generate untreated and compound-treated multi-organ states and report predicted changes in measurements such as blood glucose concentration, muscle strength or

pathological scores, together with uncertainty estimates. Relative improvement is described against normal reference ranges or predefined improvement targets. Candidate indications will then be prioritized according to the study objectives, considering the biological significance of each endpoint, the expected improvement, the evidence supporting the prediction and functional changes in other tissues. The evaluation criteria for each disease model will remain explicit so that researchers can compare candidates and select follow-up experiments. Transcriptomic data will provide biological context and suggest candidate mechanisms, while phenotypic measurements will provide the basis for evaluating function.

For example, a compound might improve metabolic abnormalities in one model and affect muscle function in another. Each prediction should specify which endpoint is expected to improve, the relevant dose and time range, accompanying changes in other tissues, and the observations or extrapolations supporting the recommendation. Researchers can then consider disease mechanisms, existing treatments and practical constraints on further development when deciding which disease models to investigate experimentally.

For disease models whose data have not yet been incorporated, PVMouse 3.0 can generate an initial state from the specified gene-editing conditions and use it to predict the response to a candidate compound. This involves extrapolation of both the disease state and the drug response, and both must be explicitly identified in the results. Exploring potential indications thus becomes a computational screening task across disease models. The purpose of this application is to narrow the set of disease models requiring full experimental evaluation and identify promising research directions beyond the original study plan.

7.4 Application 2: Screening Drug Combinations and Dosing Schedules

Within a defined disease context, a research team may need to decide which drugs to combine and how to administer them. For three candidate compounds, A, B and C, the options include each drug alone, the two-drug combinations A+B, A+C and B+C, and the three-drug combination A+B+C. For each drug, the regimen specifies the dose, start time, administration sequence, dosing interval, frequency and treatment duration. The number of drugs and the range of candidates depend on the research question. This workflow applies to gene-edited disease models already incorporated into PVMouse 3.0 or represented through extrapolation.

Multiple drugs can jointly alter a disease state through complementary actions, compensatory responses and cross-talk between tissues. The history dependence discussed in Chapter 2 means that administering A before B may produce a different state trajectory from applying them together. A regimen can therefore be represented as a combined input that varies over time. The dynamics module described in Chapter 5 predicts the resulting state trajectory, which is then translated to comparable measures of disease status and tissue function. Single-drug data provide an observational basis for each intervention, while combination and repeated-dose data help the model learn interactions and effects that depend on timing and treatment history.

The regimen description records dose, administration times, sampling times and tissue exposure separately. The complete dosing history specifies the interventions applied over time, while sampling times locate each observation within the response trajectory. For repeated dosing or sequential treatment, the model uses the dose and timing of each administration to predict subsequent changes. When drug effects are modeled in terms of tissue concentrations, exposure measurements or a pharmacokinetic module provide estimates of how those concentrations change

over time. The dynamics model links observations collected at discrete time points to predict trajectories under the specified conditions.

Regimen screening compares expected efficacy and multi-organ risks at predefined evaluation times. Disease endpoints may include functional, pathological or behavioral measurements, with objectives such as achieving a specified improvement or returning a measurement to its physiological range.

These criteria define an optimization problem: the model searches for candidate regimens within predefined sets of drug combinations and ranges of doses and schedules. A small number of discrete candidates can be evaluated individually. When dose or timing varies continuously, the search can use progressively finer grids, Bayesian optimization (*Snoek et al., 2012*) or other methods. The report documents the search space and prioritized regimens, including the dose and schedule for each drug, predicted endpoint values and multi-organ trajectories, and comparisons with single-drug treatment and the untreated disease course. Alternative regimens with different trade-offs in efficacy, effects on other organs and complexity of administration are retained for comparison. For disease models not represented in the training data, the search can begin from extrapolated states, with this basis made explicit in the resulting candidates for experimental testing.

7.5 Application 3: Identifying Multi-Tissue Risk Signals and Comparing Candidate Compounds

When several compounds offer similar expected therapeutic benefits, their effects on other tissues can influence which candidates are advanced. Drug effects can extend beyond the target organ through metabolic, circulatory, immune or neuroendocrine pathways, and the disease state can modify these responses. Using multi-tissue measurements from the same animal, PVMouse 3.0 can jointly predict treatment-related phenotypes and responses across tissues, helping researchers compare candidates and identify tissues, functional measurements and time windows for closer examination.

The model will be trained jointly on gene expression profiles and measurements of tissue function, physiology, biochemistry and pathology, represented through distinct observation channels. For example, liver gene expression can be linked to transaminase measurements and evidence of tissue injury, while cardiac states can be linked to measurements of heart function or electrophysiological activity. These relationships support predictions of both molecular changes and functional consequences. Because changes in gene expression may reflect pharmacological activity, adaptive responses to stress or injury, safety evaluation must be based on the functional and pathological measures of interest. Open TG-GATEs, which combines post-treatment gene expression profiles with conventional toxicological observations, provides a precedent for organizing data in this way (*Igarashi et al., 2015*).

Candidates can be compared by examining predicted tissue-level measurements at a predefined level of improvement in a disease endpoint, or by considering overall benefits and risks under matched study conditions. For example, if two compounds are predicted to improve the target phenotype but one also shows stronger predicted signals associated with liver injury, further liver assessment can be prioritized. Comparisons must account for dose and tissue exposure so that differences in response magnitude are not attributed solely to compound identity. The resulting

report will summarize each candidate's predicted multi-tissue effects, the evidence supporting potential risk signals and priorities for follow-up assessment.

7.6 Application 4: Screening Therapeutic Targets and Determining How to Modulate Them

Before a candidate compound has been selected, PVMouse 3.0 can use a disease-model digital twin to identify gene-level interventions that may improve predefined disease endpoints. The search can consider inhibition, partial reduction, restoration or enhancement of gene function or expression, including combinations of targets. Because a disease state may be sustained by several interconnected processes, candidate targets need not be limited to the original disease-related gene. They may also include nodes in the molecular prior network involved in downstream processes, alternative pathways or compensatory responses. The model will compare the predicted effects of these interventions on the intended functional endpoints and on other tissues.

The genetic modification used to create a disease model must be represented separately from a therapeutic intervention. A conventional knockout present throughout development differs in both history and scope from temporary, partial inhibition of the same gene after disease onset. Gene augmentation must likewise be represented in terms of where and to what extent it restores function. A MECP2 rescue study in mice with MeCP2 overexpression showed that reducing MeCP2 through genetic manipulation or antisense oligonucleotides (ASOs) improved multiple phenotypes, providing an experimental precedent for modifying a disease state by adjusting gene dosage (*Sztainberg et al., 2015*).

Candidate interventions are compared using the same disease-model construct and initial state, with each intervention specified by its target gene, direction and extent of modulation, target tissues and timing. The molecular prior network provides relationships among candidate targets, the perturbation model predicts subsequent multi-tissue states, and phenotype readout models translate these states into predicted disease endpoints and effects on other tissues. Predefined gene sets and an initial broad screen can narrow a large candidate set before modulation strength and tissue specificity are examined in greater detail. Predictions for targets or combinations not represented in the training data will be explicitly labeled as extrapolations.

The resulting report will present prioritized interventions, their predicted benefits and potential adverse effects on other tissues, and the network relationships and perturbation data supporting each candidate. These results can guide genetic rescue or functional intervention experiments. As candidates progress toward small-molecule, antisense oligonucleotide, siRNA or gene augmentation approaches, the model must also account for the actual intervention method, the tissues reached by delivery and the extent of functional modulation. This connects target selection to the practical conditions of intervention while considering both local mechanisms and whole-organism consequences.

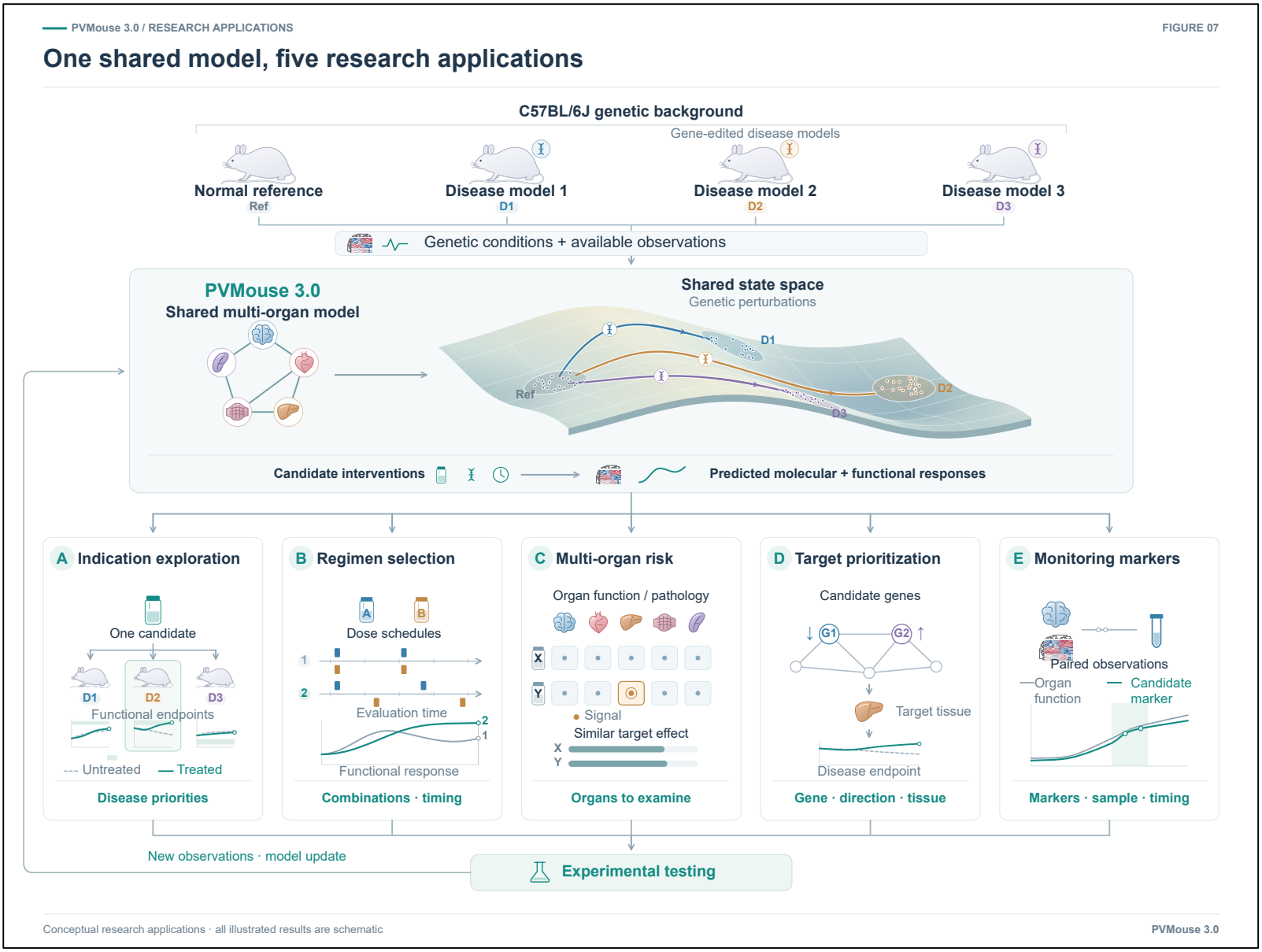
7.7 Application 5: Identifying Candidate Measures for Pharmacodynamic and Safety Monitoring

Drug development requires measurements that reflect intended pharmacological effects or organ injury. PVMouse 3.0 can jointly analyze multi-tissue states, dose and treatment timing, and functional phenotypes to identify candidate markers for these processes. A biological process may

leave related signals in different measurements: changes in tissue metabolism, for example, may be accompanied by changes in local gene expression and blood biochemistry. Linking these observations to tissue states can help identify accessible signals that reflect changes in a target tissue and determine when they appear. Candidate selection must therefore consider both the relationship to functional outcomes and the suitability of sampling and assay methods. Pharmacodynamic monitoring, safety monitoring and dose selection are among the uses described in the FDA's overview of biomarkers.

The analysis begins by defining the functional process to be monitored and using paired tissue transcriptomic and phenotypic data to identify candidate signals. Candidates are assessed by their association with target readouts, whether their changes track those readouts over time, their responses to treatment and their relationships with changes in other tissues. Tissue RNA expression must not be treated as equivalent to protein abundance in blood or urine; relationships between these measurements require paired observations or supporting models. When only tissue transcriptomic data are available, screening can identify relevant expression features and nominate molecules for further measurement. As paired peripheral data become available, the model can evaluate whether readily measured signals reflect the target organ's state and whether combining several markers provides additional information.

The workflow can use samples collected at experimental endpoints from separate animals at different time points. Where feasible, blood, urine and tissue samples can be collected from the same animal at the endpoint; invasive sampling of multiple tissues before treatment is not required. Comparisons across time points initially characterize population-level responses. To support practical monitoring studies, outputs will specify the tissue or function associated with each candidate marker, its expected direction of change, the appropriate observation window, and the proposed sample type and assay.



One shared model for five research applications. Normal reference animals and three gene-edited disease models on the C57BL/6J background use a shared multi-organ model and state space. Genetic conditions and available observations define the starting context; candidate interventions yield predicted molecular and functional responses. (A) Indication exploration compares functional endpoints across disease models, including untreated and treated states and the illustrated target range. (B) Regimen selection compares combinations, dose schedules and evaluation times. (C) Multi-organ risk assessment compares organ function or pathology at similar target effects and identifies signals for examination. (D) Target prioritization considers candidate genes, directions of modulation, target tissues and disease endpoints. (E) Monitoring-marker exploration relates paired organ-function and marker observations to sampling windows; the shaded window is a candidate sampling period. Experimental testing returns new observations for model updates. Illustrated results are schematic.

7.8 Comparing and Connecting the Five Applications

The five applications share state representations, perturbation interfaces, and mechanisms for decoding observations, but differ in which conditions are held fixed, which variables are explored, and which research and development decision the outputs support.

Application	R&D question	Main inputs	Computational task	Output
Exploring candidate indications	Which diseases warrant further study of a compound?	Compound response data and digital twins of disease-model strains	Compare treated states across disease contexts	Candidate indications and experiment priorities
Screening drug combinations and dosing schedules	How should interventions be combined for the target disease?	Disease state, drug data, regimen range, and objectives	Search joint perturbations and time courses	Preferred combinations, schedules, and alternatives
Identifying multi-tissue risk signals	Which tissues are affected, and how should candidates be selected?	Treatment conditions, multi-tissue data, and functional and pathological measures	Jointly compare therapeutic benefits and tissue risks	Compound comparisons and priority assessments
Screening therapeutic targets	Which genes should be modulated, and in which tissues?	Disease state, gene priors, and candidate interventions	Simulate interventions and rank candidates by predicted phenotypic outcomes	Targets, directions, tissues, and expected consequences
Discovering monitoring measures	Which measurements can track pharmacological effects or injury?	Paired tissue and phenotype data and available peripheral readouts	Learn cross-modal relationships and screen candidates	Measures, sample types, and observation windows

These applications can be used together to address a common research question. For example, indication exploration can identify disease models for further testing, while combination screening can help select dosing regimens. Multi-tissue risk analysis can highlight tissues and functional changes that require closer examination, and the selection of measures for monitoring drug effects and tissue injury can help determine what to measure and when. Therapeutic target identification can also suggest new intervention strategies to guide subsequent work on candidate compounds and treatment regimens.

At the PVMouse 3.0 stage, these applications are intended to identify candidates and research priorities within the mouse system. Their relevance to clinical development will need to be established through studies of human disease tissues, genetics and function, testing whether the proposed interventions and underlying biological processes translate to humans.

Chapter 8 | From Architecture to a Working System: The Development Path and Evolving Capabilities of PVMouse

Chapter 5 defined the computational architecture, training and inference mechanisms, and validation principles of PVMouse 3.0, while Chapters 6 and 7 discussed continual learning and research applications. Building on the available data and model components, this chapter explains where development will begin, how successive stages depend on one another, and how the complete system will be used and tested in research.

8.1 The Scope and Starting Point of PVMouse

PVMouse 3.0 starts with C57BL/6J mice, initially covering the brain, heart, liver, skeletal muscle, and spleen. Given the genetic background, sex, age, perturbation conditions, and prediction window, it aims to jointly predict transcriptomic changes in these five tissues and organs, associated physiological and pathological readouts, and changes in state over the selected period. Single-gene knockouts and other genetic perturbations will provide the first modeling and testing tasks; capabilities for predicting responses to drugs, diet, and environmental stimuli will develop as the relevant data are incorporated, the model is adapted to these tasks, and its predictions are independently validated.

The project has built some algorithmic components and is cleaning the collected data while continuing to collect and organize knockout-related resources. The overall system remains to be completed and validated.

As of October 25, 2026, the project has obtained bulk RNA-seq data for 4,741 tissue samples from C57BL/6J mice, spanning 1–116 weeks of age and covering the brain, heart, liver, skeletal muscle, and spleen, with links retained between tissues from the same animal. Bulk RNA-seq data have also been obtained for a further 2,511 tissue samples from mice subjected to genetic or other perturbations on the C57BL/6J background, likewise retaining the relationships among tissues from the same animal.

Within these two collections, 775 brain bulk RNA-seq tissue samples spanning multiple ages have been identified for the first spatial organ prototype. This subset includes samples from normal C57BL/6J mice and from knockout mice on the same genetic background. It includes samples collected at ages matching those used in public single-cell transcriptomic and single-cell spatial transcriptomic atlases, providing a basis for organ modeling with age-matched cellular and spatial references. All bulk RNA-seq counts above refer to tissue samples, with different tissues from the same animal counted separately.

Development will proceed in the following order: first establish the brain spatial prototype and the other four tissue and organ modules; then build the normal C57BL/6J joint five-organ state; and finally integrate genetic perturbations and temporal modeling to complete PVMouse 3.0. Each stage will establish reusable state and observation interfaces, allowing subsequent training to build system-level relationships and response capabilities on existing modules and adapt them as new observations become available.

8.2 Reusable Data, Models, and Core Components

Development begins by standardizing gene and sample identifiers and recording animal backgrounds, anatomical sampling regions, and sampling times so that data from different sources can be used together within the computational framework of Chapter 5. Public single-cell and spatial transcriptomes provide references for organ modeling, while tissue transcriptomes and functional readouts from the target animals calibrate model estimates. Observations from multiple tissues of the same animal further support learning of relationships between organs. The roles of these resources and their integration into the model are summarized below.

Resource or component	**Role in development**	**Data integration and connection to the model**
mouse molecular knowledge graph	Build the molecular prior layer and provide structural and functional priors for cell state learning.	Integrate biomolecular interactions, pathways, gene sets, transcription factor-target relationships, and protein complexes. Standardize entity identifiers, retain relation types and evidence sources, and use graph encoding to produce prior representations for model use.
Single-cell transcriptomic foundation models	Provide adaptable cell state representations as inputs to organ modeling.	Adapt models to mouse genes, data context, and target tasks. Select and update components according to downstream prediction performance.
Single-cell or single-nucleus transcriptomic atlases across ages, including Tabula Muris Senis (Tabula Muris Consortium, 2020)	Establish references for cellular states and composition across ages and tissues.	Organize reference data by strain, age, sex, and tissue conditions. Combine spatial, bulk, and functional observations to calibrate cellular states and composition in virtual organs.
Mouse brain spatial atlases, including the Allen MERFISH-C57BL6J-638850 data associated with Yao et al.'s whole-brain atlas, as well as Zhang et al.'s whole-brain atlas and Han et al.'s whole-brain atlas	Provide references for regional cell type distributions, spatial positions, and neighborhoods in the brain.	Align cellular states with anatomical regions, spatial expression, and local composition to constrain virtual brain organization. For each dataset, retain the source strain and age information and distinguish measured data from inferred information. Other organ modules use their available spatial and anatomical resources.
Multi-tissue bulk RNA-seq across ages, including the multi-tissue aging atlas, Schaum et al.'s study, and GSE132040	Calibrate overall organ expression, learn joint multi-organ states, and establish age-related population-level physiological references.	Link tissue samples from the same animal through animal identifiers. Record the anatomical regions sampled, ages, and experimental backgrounds in each study to select comparable data. Paired observations support learning of organ relationships; observations across ages describe population state distributions and their changes.
Deep phenotyping of male C57BL/6J mice across ages: Xie et al.'s study	Provide age-related physiological, biochemical, pathological, and functional references.	Use cross-sectional observations at 3, 5, 8, 14, 20, and 26 months to establish phenotype references. Expression and phenotype measurements linked to the same animal can further support learning of molecular-functional relationships.

SLAM longitudinal phenotype resources: study design and methods	Constrain and test predictions of functional changes over time.	Select C57BL/6J cohorts from the data actually obtained. Retain animal identity, sex, repeated-measurement times, and assay conditions, and compare longitudinal changes with predictions.
ARCHS4 and other public genetic perturbation data	Learn tissue responses to genetic changes and test perturbation prediction.	Select data with clearly defined genetic backgrounds, editing constructs, controls, and sampling conditions. Combine available single-cell and spatial references to establish perturbation-response tasks, retaining study sources and actual pairing.
kallisto	Support expression quantification and consistent preprocessing of raw RNA-seq data.	Use consistent references and quantification workflows for raw sequencing data requiring reprocessing. Connect existing expression matrices according to their processing methods and sources.
Tangram (*Biancalani et al., 2021*)	Support alignment of single-cell or single-nucleus expression references with spatial observations.	Learn mappings between expression and spatial positions within the corresponding tissue regions, providing spatial constraints for cell-region-organ modeling.
Modules for organ-state aggregation and prediction of observable readouts to be developed or integrated	Organize cellular and regional states into organ representations and predict overall expression and functional readouts.	Build organ states from cell-type composition, local structure, and observation aggregation relationships, and train jointly through bulk, spatial, and phenotype prediction tasks.
Organ network and neural-network-based dynamics modules to be developed or integrated	Learn relationships among organs and how their states change over time and under interventions.	Represent organs as nodes with organ states as node features, and use graph message passing to model relationships among organs. Learn dynamic relationships using age, treatment history, and perturbation conditions, and test them with longitudinal observations and controlled experiments.

The project's own data from normal mice, knockout mice, and the corresponding genetic controls are organized according to the same standards as public data, recording animal backgrounds, experimental conditions, and measurement methods, and identifying which tissue and phenotype observations come from the same animal. These paired observations support learning of multi-organ and molecular-functional relationships, whereas unpaired data can still inform organ references, train organ modules, or characterize responses to specific perturbations. The data available at each stage determine which training and validation tasks can be undertaken and which additional observations are needed.

Component selection follows the adaptation and comparison principles described in Chapter 5. At each training stage, the molecular prior version can be held fixed and the reference encoder backbone can be frozen, concentrating development on spatial organ generation and aggregation, cross-organ connections, state updates, and observation readouts, with adaptation where needed to connect upstream and downstream components. Specific foundation models, spatial alignment methods, and dynamic models are selected according to task performance, while established organ interfaces remain available for reuse by subsequent modules.

Compared with collecting all required data and training every component from scratch, reusing existing data and models is expected to reduce the resources needed to begin development. Cell prototypes and regional summaries will initially limit computational demands, allowing new development and experimental resources to focus on connections not yet established between components and on the key tests of whether the design works.

8.3 From a Brain Spatial Prototype to the Complete PVMouse

8.3.1 Brain Spatial Prototype: Establishing Organ Modeling Interfaces

The brain is the first spatial organ prototype because publicly available cellular, spatial, and anatomical references support relatively well-defined regional organization. Allen MERFISH-C57BL6J-638850 and the Spatial mouse brain atlas (*Han et al., 2025*) provide cell distribution and spatial expression references for C57BL/6J brains at specific ages. Combined with age-matched brain single-cell transcriptomes and bulk data from corresponding anatomical regions, these resources can support a progression from cellular states and composition, through regional organization, to organ states and measurable readouts.

Prototype development begins by aligning cellular references with brain-region coordinates and incorporating regional and neighborhood information to establish regional states and expression readouts. Spatial measurements or brain-region samples constrain predictions for the corresponding regions, whose states are then aggregated into an organ state used to predict overall expression within the modeled tissue. Comparison with measured bulk RNA-seq requires expression to be aggregated over the actual anatomical sampling area, accounting for cell composition, RNA contributions, and measurement conditions. Brain-region samples, spatial sections, and whole-brain samples therefore each retain their own anatomical scope.

Existing cell state reference components can provide a starting point for connecting cellular references, regional organization, organ aggregation, and observation readouts. Together, these connections would form a brain prototype capable of predicting regional expression, overall organ expression, and associated readouts.

To limit computational demands, representative cell states can initially summarize similar cells, while regional representations aggregate information about cell composition and state within each region. Additional cell states or finer spatial resolution can be introduced if independent observations show that particular cellular or local structural differences affect prediction. The resulting interfaces between cellular references, spatial regions, organ states, and observation readouts can then be reused by other organ modules.

8.3.2 The Other Four Organ Modules: Reusing Interfaces and Adding Their Structures

After the brain prototype is established, the same state, aggregation, and readout interfaces extend to heart, liver, skeletal muscle, and spleen. Each module incorporates its own cellular and tissue data and retains task-relevant structures: anatomical divisions of the heart, regional specialization in the liver, sampling location and muscle fiber composition in skeletal muscle, and immune microenvironments in the spleen. Organs with limited spatial data can initially be modeled at regional scales supported by existing evidence.

The four modules use consistent identifiers and representations for genes, conditions, and observations, each supplying a state summary to the joint model while retaining its own cellular

and regional information. Changes in the system state can subsequently be passed back through the same interfaces to update organ readouts. This establishes a common computational basis for joint modeling of all five organs.

8.3.3 Normal Joint States: Connecting Five Organs through Same-Animal Observations

Once the five modules are established, multi-tissue expression and accompanying physiological, biochemical, and pathological observations from the same normal C57BL/6J animals will be used to train a joint state model from which these readouts can be predicted together. Organ modules provide state summaries, with the organ graph and measurable systemic mediators organizing information transfer between them. Blood measurements and organism-level phenotypes also contribute to training, encouraging the joint state to retain information relevant to functional differences.

This stage moves from describing five organs separately to estimating their joint relationships using information from the same individual. Data with recorded same-animal pairing constrain these relationships, while other resources supplement each organ's reference distribution. Observations from populations of different ages can already establish age-conditioned normal references at this stage, providing a basis for subsequently modeling how states change over time.

This stage yields a normal C57BL/6J joint five-organ state with gene expression and phenotype readouts, allowing measurements from some tissues to inform estimates for other tissues or unmeasured phenotypes. It also establishes a shared reference for comparing systemic responses to changes in genetic conditions.

8.3.4 Genetic Perturbations and Time: Forming the Complete MVP

After the normal joint state has been established, single-gene knockouts and other genetic perturbations on the C57BL/6J background will be incorporated with appropriate controls to learn how genetic changes relate to multi-organ expression and function. The target gene, editing method, affected tissues or cells, and time at which the perturbation takes effect enter through the interface defined in Chapter 5. Differences between knockout mice and their matched controls train response prediction while adapting the relevant organ modules and relationships across the system.

Temporal modeling builds on the age-conditioned population references established earlier, combining observations across ages, samples at different times after perturbation, and available longitudinal readouts to learn how states change under normal and perturbed conditions. The first temporal task will focus on a single-gene knockout model on the C57BL/6J background and its matched controls sampled at multiple ages or post-perturbation time points, with the aim of predicting multi-organ gene expression changes and related functional readouts within a predefined period. Starting from a distribution of states constrained by the initial age, genetic construct, and observations available at that time, the model advances under explicit assumptions about dynamics and the persistent genetic condition to predict responses relative to matched controls at the target time. The starting point and treatment of genetic conditions follow the definitions in Chapter 5.

Samples from independent animals at different time points constrain and test population-level changes in response, whereas repeated functional measurements from the same animal test predictions of that individual's subsequent changes. Temporal predictions will be evaluated at

target time points held out in advance or against independent time series, and compared with age-conditioned static models and simple temporal models using the same inputs to determine whether state updates provide additional predictive information. Further observations over time and independent experiments will calibrate the candidate trajectories. This first temporal task evaluates prediction at unobserved times for the selected knockout model; generalization to unseen genetic constructs or target genes is evaluated separately.

The first complete MVP links genetic conditions to the joint five-organ state, models how that state changes over time, and predicts functional readouts from the resulting states. For selected genetic constructs and time ranges, it generates multi-organ response and associated phenotype predictions for comparison with matched controls. Finer temporal resolution, broader perturbation coverage, and analysis of propagation relationships can develop as corresponding data and experimental evidence accumulate.

These development dependencies can be summarized as follows:

Development stage	**Main work and resources**	**Resulting capability**
Consistent data and reference coordinates	Organize gene, animal, tissue, and time identifiers; integrate molecular priors and cellular references.	Reusable reference components and condition and observation interfaces.
Brain spatial prototype	Combine brain cell atlases, spatial and anatomical references, and tissue expression; develop regional and organ aggregation.	Brain regional states, organ expression, and related readouts.
Modules for the other four organs	Reuse interfaces and add structures and observations for heart, liver, skeletal muscle, and spleen.	Five-organ states and readouts that can be used together.
Normal C57BL/6J joint states	Use normal same-animal multi-tissue and phenotype data to establish age-conditioned joint references.	Joint estimation across five normal organs and organism-level phenotype readouts.
Genetic perturbations and time	Add genetic constructs, appropriate controls, and observations at different time points; train response-prediction and state-update models.	Five-organ perturbation responses, prediction of changes over time, and phenotype prediction, forming the complete MVP.

Functional and pathological readouts can contribute to training at any stage where corresponding data are available. Shared interfaces between stages allow new observations to inform existing modules, refining cellular references, spatial organization, and relationships between organs.

8.4 Research Use and Key Tests of the Complete MVP

Consider a single-gene knockout study on the C57BL/6J background. Researchers specify the genetic construct, the tissues or cells it affects and when it takes effect, the animal's sex and starting age, and the target prediction time or window. The model estimates an initial state distribution from these conditions and reference data, together with any measurements already available from the target animal, such as blood or functional readouts. This distribution reflects the effects of any genetic alteration already active at the starting point. The model then predicts gene expression and associated phenotypes in the brain, heart, liver, skeletal muscle, and spleen while that alteration remains present. Corresponding predictions for the genetic control at matched ages and experimental conditions allow knockout-associated expression and functional changes to be compared and reported separately, each with its uncertainty. Cell populations, tissue regions, and expression features associated with the predictions are presented as interpretive clues.

This use of the model supports two distinct tasks. Estimation of missing observations uses measured tissue or phenotype data from the same animal to infer unmeasured results, whereas prediction of intervention outcomes uses the genetic conditions, treatment history, and observations known at query time to predict subsequent molecular and functional changes. Inputs, target readouts, and evaluation results are specified separately for each task.

Evaluation follows the data splits and comparison methods in Chapter 5, focusing on whether the model predicts knockout-associated expression and functional changes relative to matched controls—their direction, magnitude, and differences across organs—and whether multiscale information improves predictions over simple baselines. Associated cell populations, coordinated expression changes among functionally related genes, and relationships between organs provide candidate leads for subsequent mechanistic experiments.

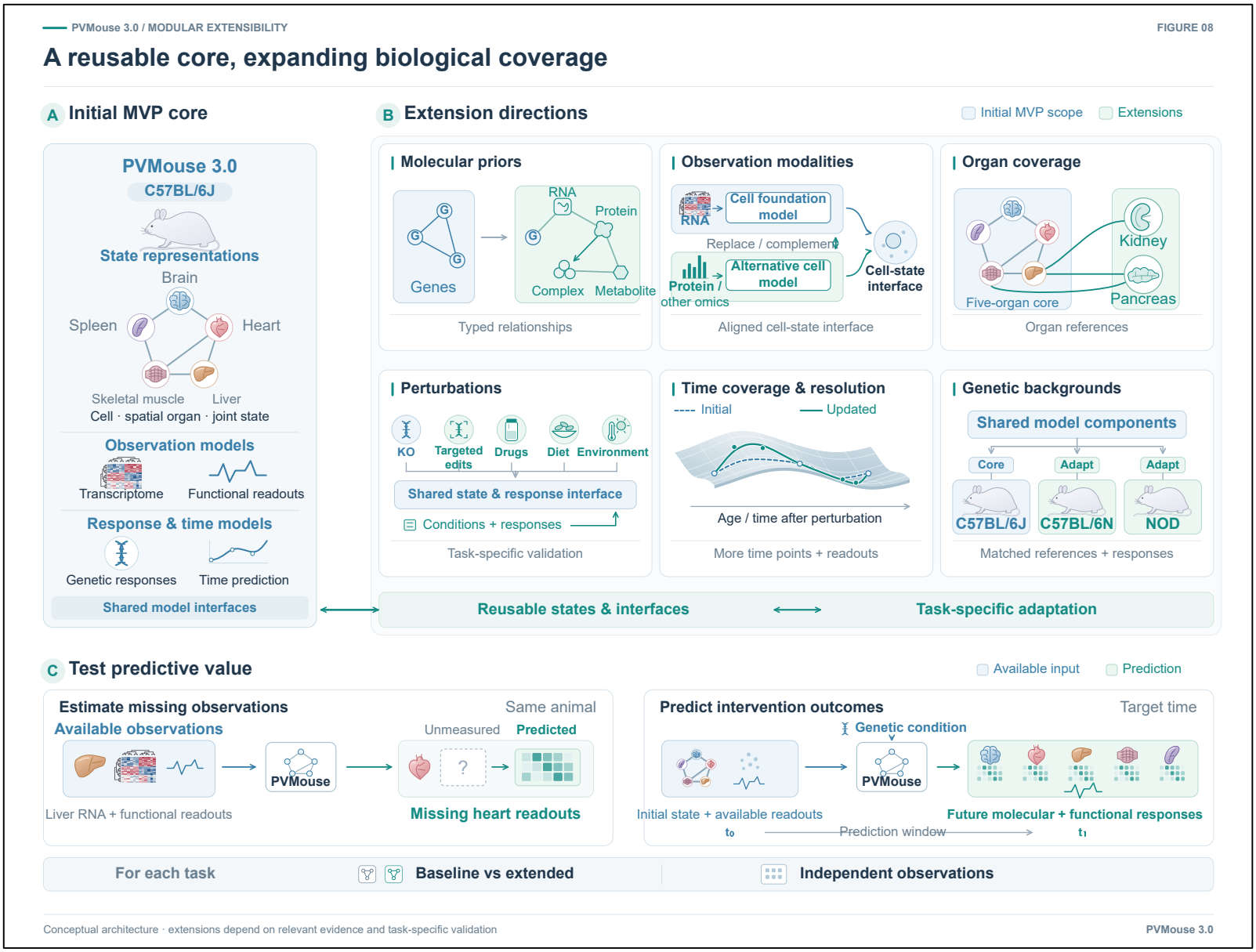
Temporal predictions will be tested using time points or longitudinal observations excluded from fitting, and compared with age-conditioned static models and simple temporal models given the same available inputs. If state updates do not improve predictions at later time points, the errors will guide examination of the initial state, temporal information, and dynamics module to determine which computations need revision or which observations need to be added.

Prediction errors and competing biological explanations guide the choice of additional observations and experiments, as illustrated by the tissue-specific perturbation example in Section 6.3. The resulting data contribute to model updates through the feedback process described in Chapter 6.

8.5 Later Extensions and Their Additional Dependencies

Once PVMouse 3.0 is established, further development will first examine whether the state representations and response models learned from genetic perturbations can support studies of drug treatment, diet, and environmental stimuli. These tasks can reuse the state and observation interfaces while incorporating their own intervention conditions, response data, and functional endpoints. Drug studies require doses, dosing schedules, and available exposure information, while dietary and environmental studies require records of the treatment, its intensity, duration, and history. Task adaptation and independent validation will establish which representations and modules can be reused and which response relationships must be learned anew, gradually extending prediction beyond genetic perturbations.

The biological extensions described in Section 4.7 will be guided by specific research questions, limitations in current predictions, and the availability of new data. Priorities will depend on the improvement that additional information brings to the target task and the costs of acquiring the data and adapting the model. Finer temporal resolution requires observations with well-defined starting points and appropriate sampling intervals, while extension to new genetic backgrounds requires corresponding normal references and perturbation data. New organs and observation modalities will likewise be incorporated around defined tasks and evaluated for their contribution beyond the existing model.



A reusable core and extensions of biological coverage. (A) The initial C57BL/6J MVP connects cellular, spatial-organ and joint-state representations with observation, genetic-response and time models through shared interfaces. (B) Six extension directions retain the distinction between the initial scope and extensions. Molecular priors add mapped entities and typed relationships, including RNA, proteins, protein complexes and metabolites; the protein-to-complex link denotes

membership. Protein or other-omic cell models may replace or complement the transcriptomic model through suitable models and paired-data alignment at the cell-state interface. Organ expansion requires organ references and joint observations. Different perturbations enter the shared state and response interface with their own conditions, response data, treatment-specific learning and validation. Added time points and repeatable readouts support updated temporal predictions. Genetic-background adaptation uses shared components together with matched references and response data for C57BL/6J, C57BL/6N and NOD. (C) Same-animal estimation of missing observations is distinguished from prediction of future molecular and functional responses under a genetic condition and prediction window. Baseline and extended models are compared using independent observations. Extensions depend on relevant evidence and task-specific validation.

Chapter 9 | Outlook and Conclusions

9.1 A Research Path from a Virtual Mouse to a Virtual Human

The long-term goal of PVMouse 3.0 is to extend validated modeling approaches gradually to humans. Human studies have accumulated extensive clinical records, blood-based measurements and imaging data, alongside more limited tissue data. However, obtaining repeated measurements of multiple biological features across organs in the same person under controlled conditions remains difficult. Differences in genetic background, living environment, disease history, and treatment experience further complicate state estimation and response prediction. The mouse system therefore provides a foundation for developing and testing approaches that link local measurements to whole-organism phenotypes and responses to controlled interventions.

Progress toward a virtual human could build on state representations, candidate biological relationships and model components developed in mice, with their applicability assessed using human data. Initial work could focus on specific organ processes, disease contexts and prediction tasks, gradually developing models of human biological states and how they change. A central question is how repeated peripheral measurements from the same person can help infer tissue states that are difficult to observe directly.

9.2 Connecting Blood Observations with Tissue States

Blood provides relatively accessible material for repeated sampling in longitudinal human studies. Biochemical measurements, blood cell profiles and circulating molecules provide information about different aspects of physiological change. Plasma extracellular vesicles (EVs) originate from diverse cell types. Their membrane components, together with their RNA, proteins, lipids and any associated DNA, may provide further information about biological state. Interpreting these measurements requires appropriate EV separation and characterization to distinguish material within vesicles from material associated with their surfaces or co-isolated during sample preparation, following the MISEV2023 recommendations (*Welsh et al., 2024*).

EVs also have measurable biophysical properties. *In vitro* studies have defined nanomechanical differences between small EVs released by senescent and non-senescent cells (*Lee et al., 2022*), while high-speed atomic force microscopy (HS-AFM) has been used to examine changes in EV morphology under different physicochemical conditions (*Sajidah et al., 2022*). These studies provide a basis for jointly analyzing EV molecular composition, morphology and mechanical responses. Whether this approach can help assess human tissue states and aging remains to be tested by pairing EV measurements with tissue assessments and aging-related observations.

PVMouse 3.0 can provide an experimental and modeling basis for establishing these connections. Pairing plasma EV features with tissue expression, pathology, and physiological measures from the same animal across ages and controlled perturbations can reveal which peripheral signals relate to particular tissue processes. For example, if a liver perturbation changes both tissue metabolism and plasma EV features, later studies can test whether those EV features help assess liver state and how much information they add beyond conventional blood measures. Tissue-specific perturbations, suitable source tracing, and independent experiments can further constrain the interpretation of these relationships.

Changes in plasma EV measurements may reflect not only tissue states but also shifts in EV population composition, release and clearance. Relationships identified in mice must therefore be tested and, where necessary, adapted using longitudinal human blood measurements, clinical indicators, imaging and available tissue data. If validated, EV features could complement other observations in estimating human biological states and predicting subsequent changes.

9.3 Data Infrastructure for Cross-Species Research

Moving from a virtual mouse toward a virtual human requires sustained sample and data collection guided by shared research questions. Mouse studies can link blood measurements to observations from multiple tissue omics and whole-organism phenotypes. Human studies can relate repeated blood measurements to disease progression, treatment history, imaging and functional assessments. Together, these resources can help determine which state features transfer across species and which relationships need to be learned again from human data.

Mouse studies could further develop standardized procedures for collecting and storing samples from multiple tissues. At planned experimental endpoints, blood and tissue sampling could be coordinated to meet both current study requirements and anticipated future assay needs, while preserving the link between each sample and its source animal. Machine vision, robotic dissection and sample-tracking systems could eventually assist with tissue identification, collection and labeling. These tools could improve consistency and help maintain records of sampling procedures and sample quality.

Universities, research institutions, contract research organizations, and pharmaceutical companies could build on this approach to explore a shared network of mouse tissue samples. Participating institutions would register samples available for further research, allowing other researchers to search and apply for access according to the experimental background, tissue type, and assay requirements. Samples held across institutions could thereby support new research questions.

Sample sharing must also preserve the connection between new measurements and the original experiment. Consistent identifiers for animals and their tissue samples would allow results from subsequent analyses to be linked to the correct animal and experimental records, under agreed rules for access and acknowledgment of contributions. Measurements generated by different teams could then contribute to increasingly comprehensive datasets for individual animals. Where stored samples remain suitable for additional assays, their reuse could reduce the need for further animal experiments.

This infrastructure could begin with standardized manual collection, sample registration, and small-scale sharing, introducing automation and broader collaboration as needs arise. Automated sampling and cross-institutional sharing would be evaluated progressively in terms of sample quality, efficiency of use, and research benefits. The resulting resources, linking blood and tissue measurements from the same animals, would support studies of how EV features relate to organ states. Comparison with longitudinal human data would then help assess how these findings could contribute to models of human biological states.

9.4 Whole-Body Spatial Observations and Unified Modeling

Whole-body spatial transcriptomics offers another long-term direction for biological world models. By combining Array-seq with histological imaging of whole-body sagittal mouse sections, *Clevenger et al., 2026* showed how molecular states and tissue structures can be characterized together across organs in a shared anatomical context. This work provides a technical foundation for developing more systematic whole-body atlases. With sufficient resources, systematic sampling and reference coordinate construction similar to those used in the Allen brain atlas could be extended across multiple ages and controlled perturbations. Serial whole-body sections, spatial transcriptomic measurements and single-cell reference data could then be integrated with physiological observations to build a reference atlas of biological states across space and time.

As measurement resolution improves and coverage expands across anatomical regions and experimental conditions, researchers could explore a unified model of whole-body biological states. This model could use single-cell foundation models to learn representations that incorporate spatial location, relationships between regions, and information about time and interventions. It could learn how local information is integrated, how effects propagate over longer distances and how biological states relate to functional measurements. This could reduce reliance on reconstructing virtual organs separately and specifying connections between them in advance. However, compared with reusing public datasets collected across separate studies, this approach would require substantially greater investment in sample collection, experimental measurements and computation. Its predictive value would still need to be tested using independent observations and intervention experiments. The current modular approach could contribute reusable state representations and interfaces, along with validation tasks, while new whole-body data could progressively refine and extend the existing model.

9.5 Summary

Understanding complex disease and intervention outcomes requires linking local molecular changes to function at the cellular, tissue and organism levels. Guided by working hypotheses concerning constrained dynamics, multiscale organization, history dependence and macroscopic predictability, this white paper proposes using a biological state space to organize observations from separate studies and interconnected models to describe biological states and how they change. This approach brings observations from different levels together to inform predictions and links model-based explanations to specific experimental findings.

PVMouse 3.0 begins with five organs and tissues in C57BL/6J mice, combining a normal physiological reference, genetic perturbations, and paired molecular and functional observations from the same animal. Development and validation will proceed in stages. **The central test is whether incorporating cellular and spatial information, together with information from other tissues, enables more accurate predictions of gene expression changes in a target tissue and associated functional outcomes than simpler models can provide.** Comparing predictions with experimental results will help determine which information sources and model components improve prediction. These comparisons will also identify gaps in the observations, guide revisions to modeled relationships and inform the choice of subsequent experiments.

Data organization and the development of some algorithmic components are underway, but the complete system remains to be built and tested. By sharing the scientific rationale and technical design, this white paper aims to encourage discussion and collaboration on modeling methods, data resources and joint experiments. As more observations and validation evidence become

available, the model can be extended to additional interventions, tissues and biological systems, informing future efforts to adapt the approach to humans. Its long-term value lies in enabling knowledge from different studies to be continually incorporated into a shared computational framework in which new predictions are tested experimentally. Through this process, findings from individual studies can build toward a more complete understanding of living systems.

Acknowledgement and AI assistance

The core research ideas presented in this white paper—including the four working hypotheses for biological modeling, the biological and computational architectures of PVMouse 3.0, and its staged development strategy—were conceived and developed by the authors. OpenAI's ChatGPT and Codex assisted with iterative discussion and critical examination of these ideas, manuscript organization, English translation and revision, and figure preparation. The authors evaluated the AI-generated suggestions and made the final scientific and editorial decisions, and take responsibility for the claims, interpretations, and final content of this white paper.

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